

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

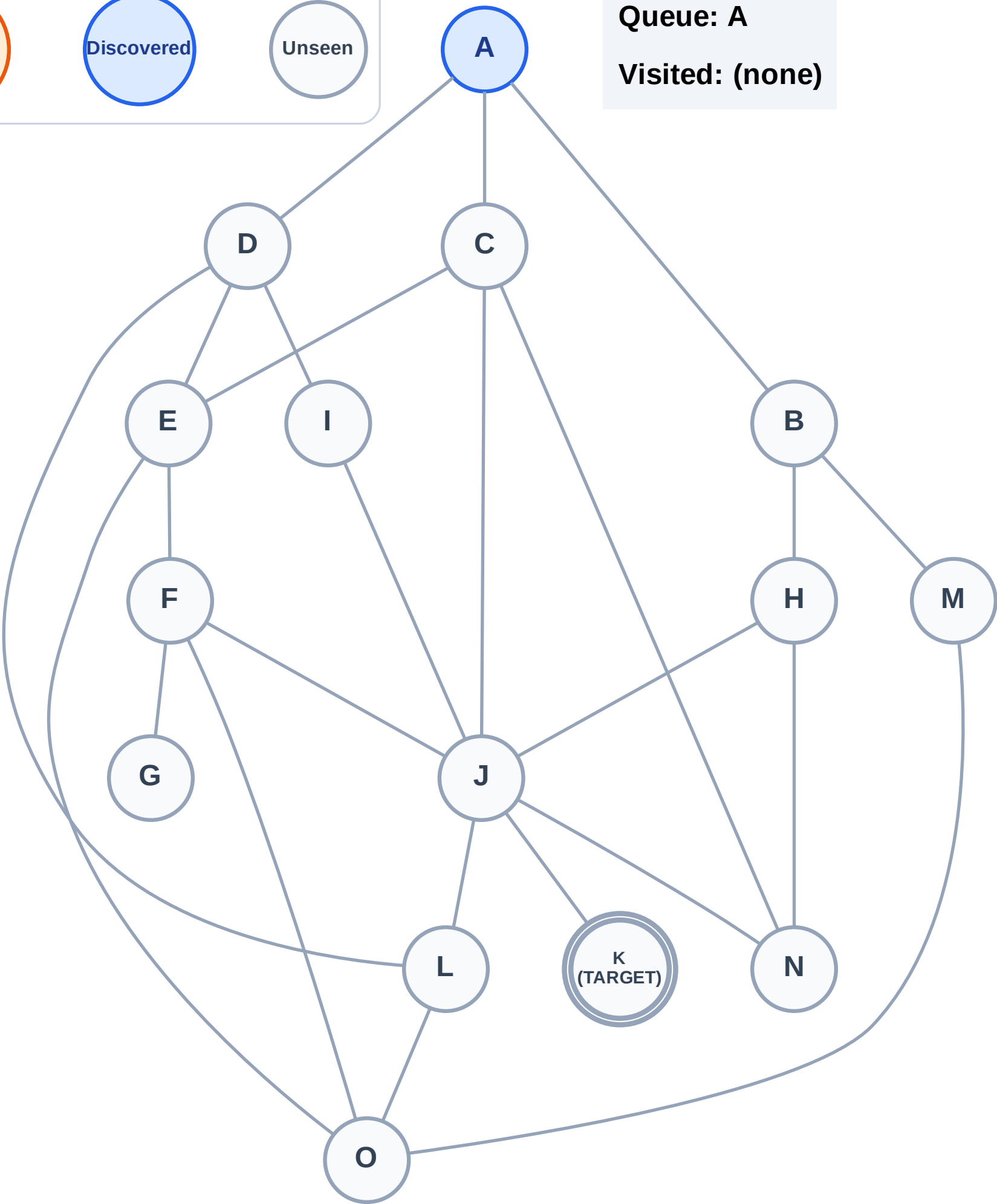
Complete

Processing

Discovered

Unseen

Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

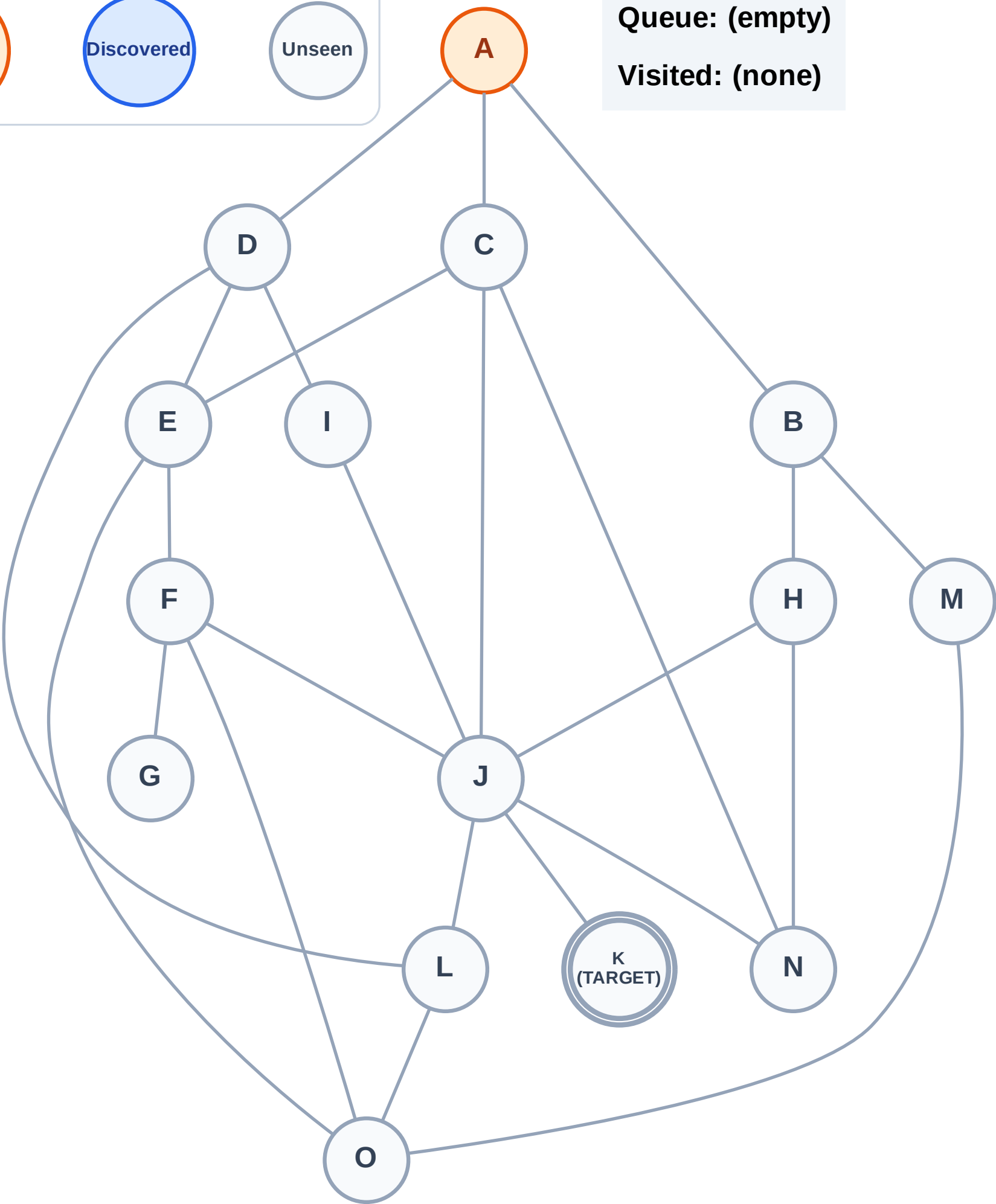
Complete

Processing

Discovered

Unseen

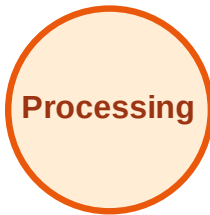
Queue: (empty)
Visited: (none)



BFS Traversal

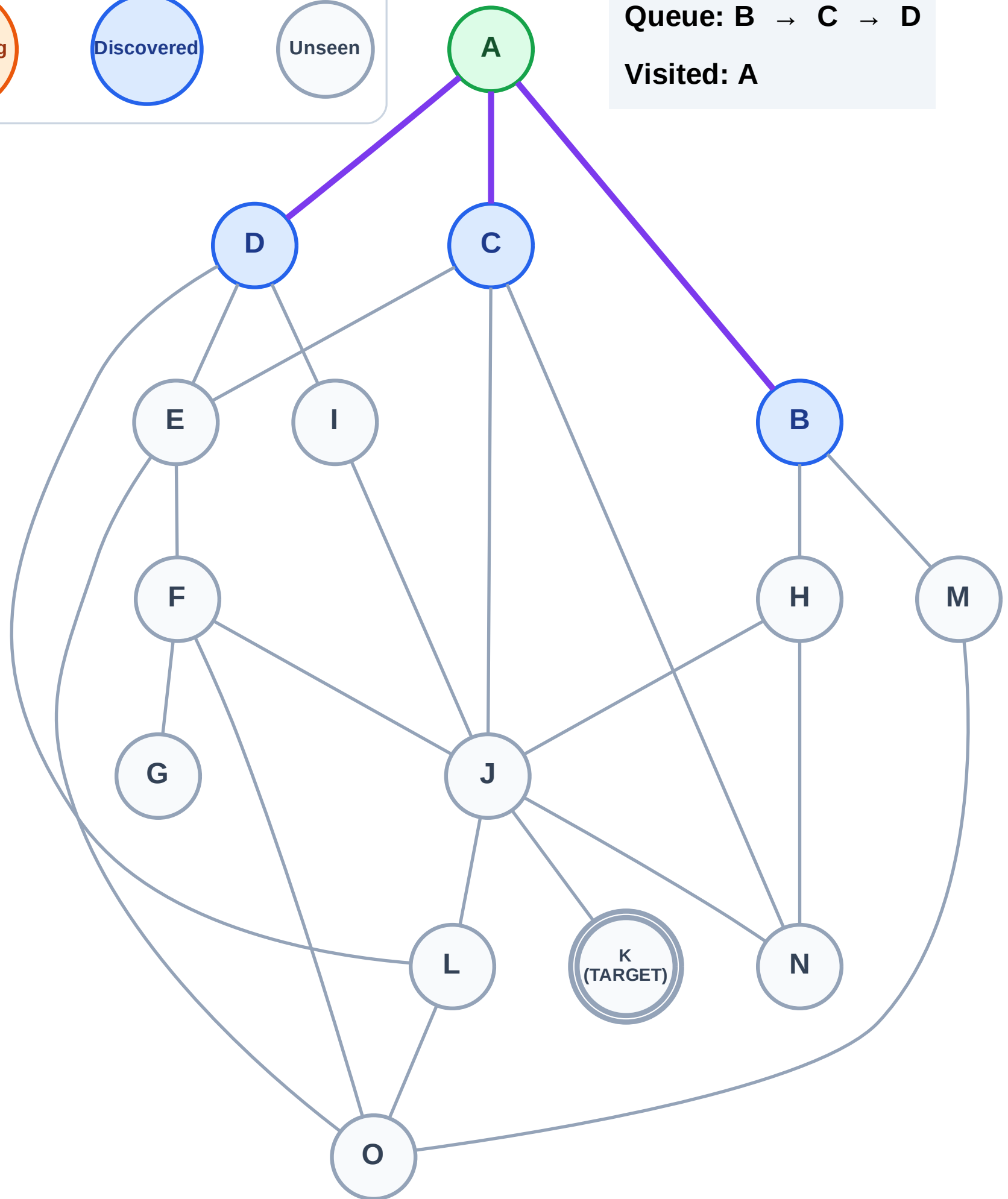
Step 3: Discover B, C, D from A

Legend



Queue: B → C → D

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

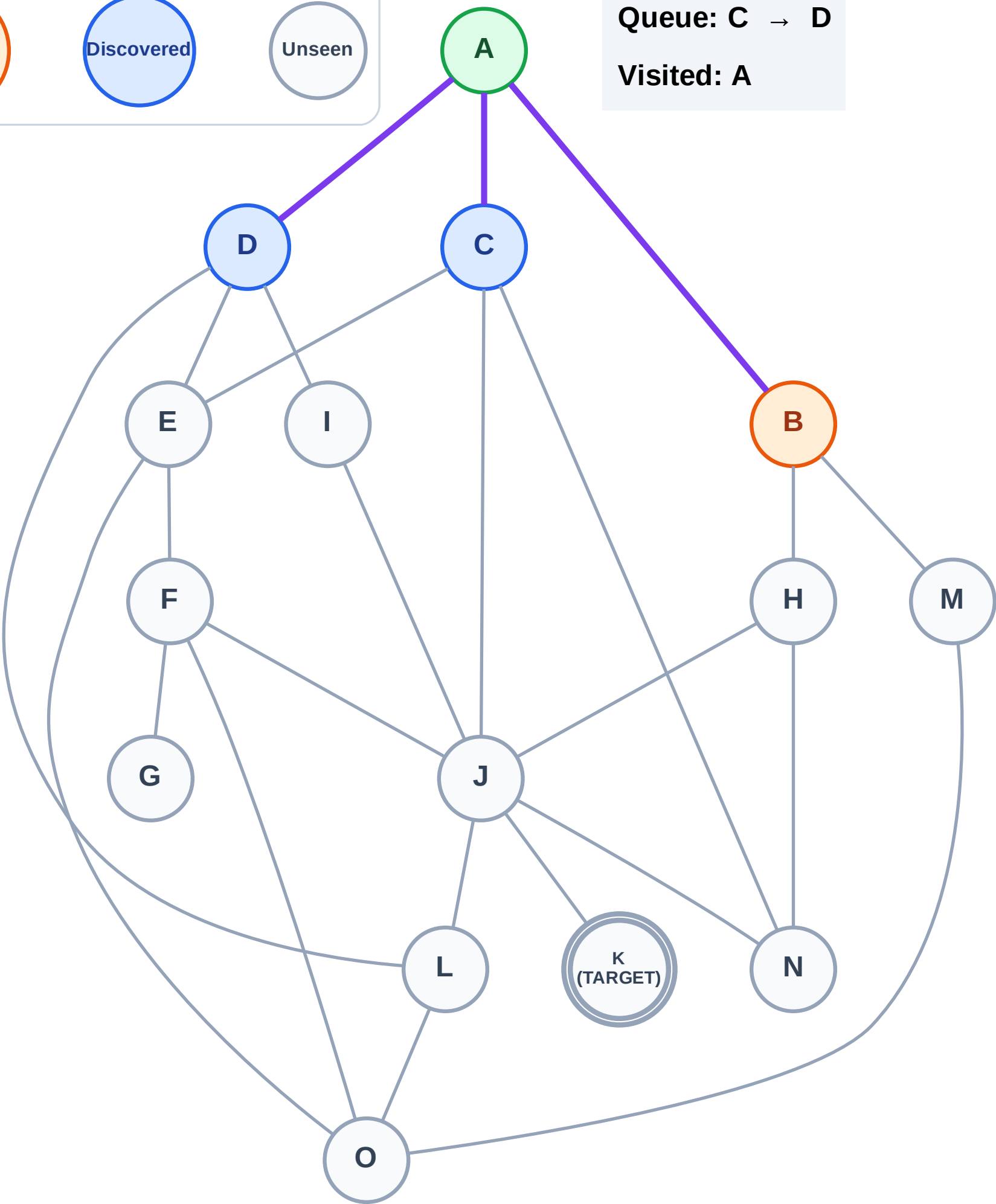
Processing

Discovered

Unseen

Queue: C → D

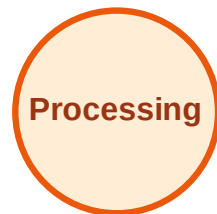
Visited: A



BFS Traversal

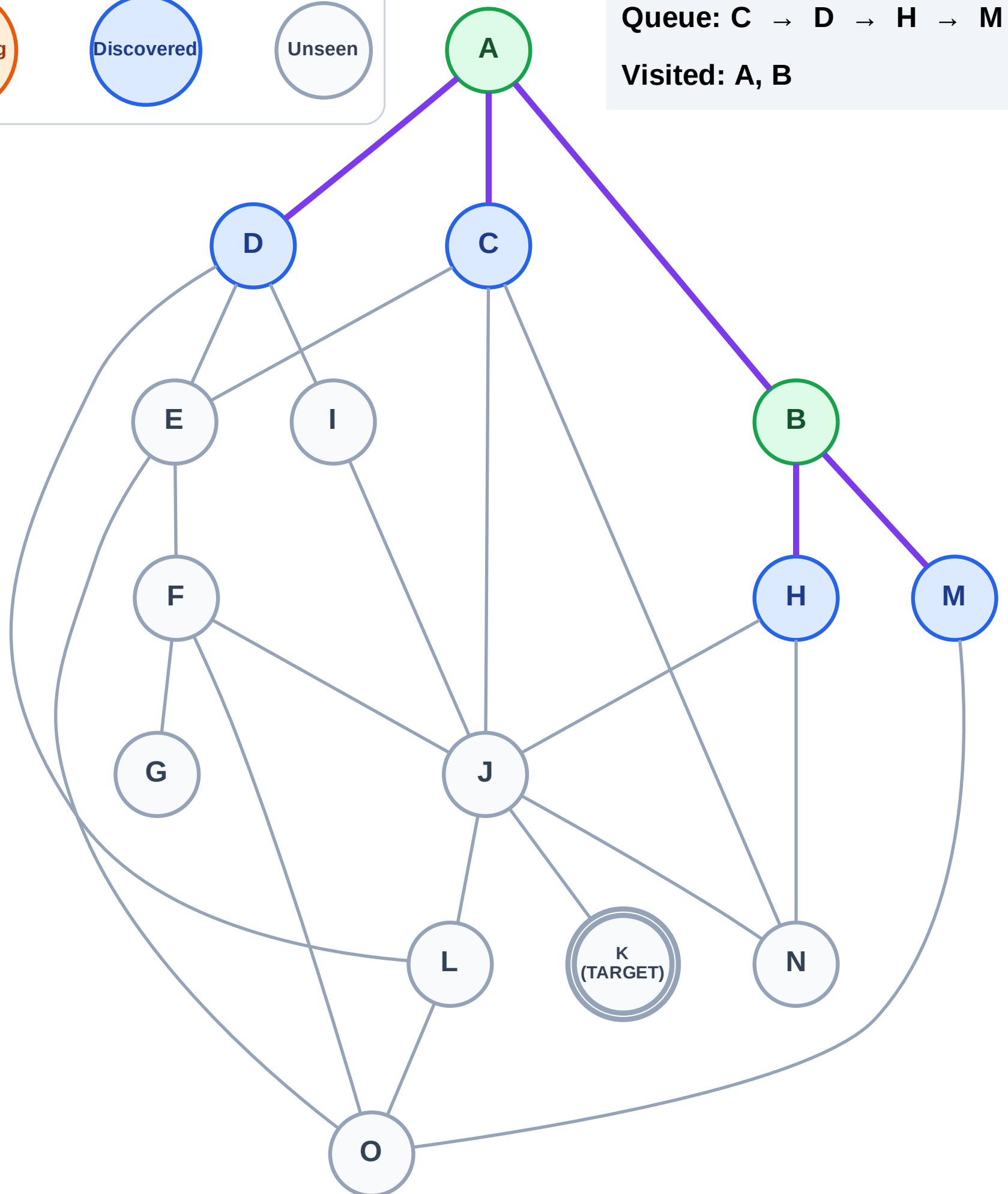
Step 5: Discover H, M from B

Legend



Queue: C → D → H → M

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

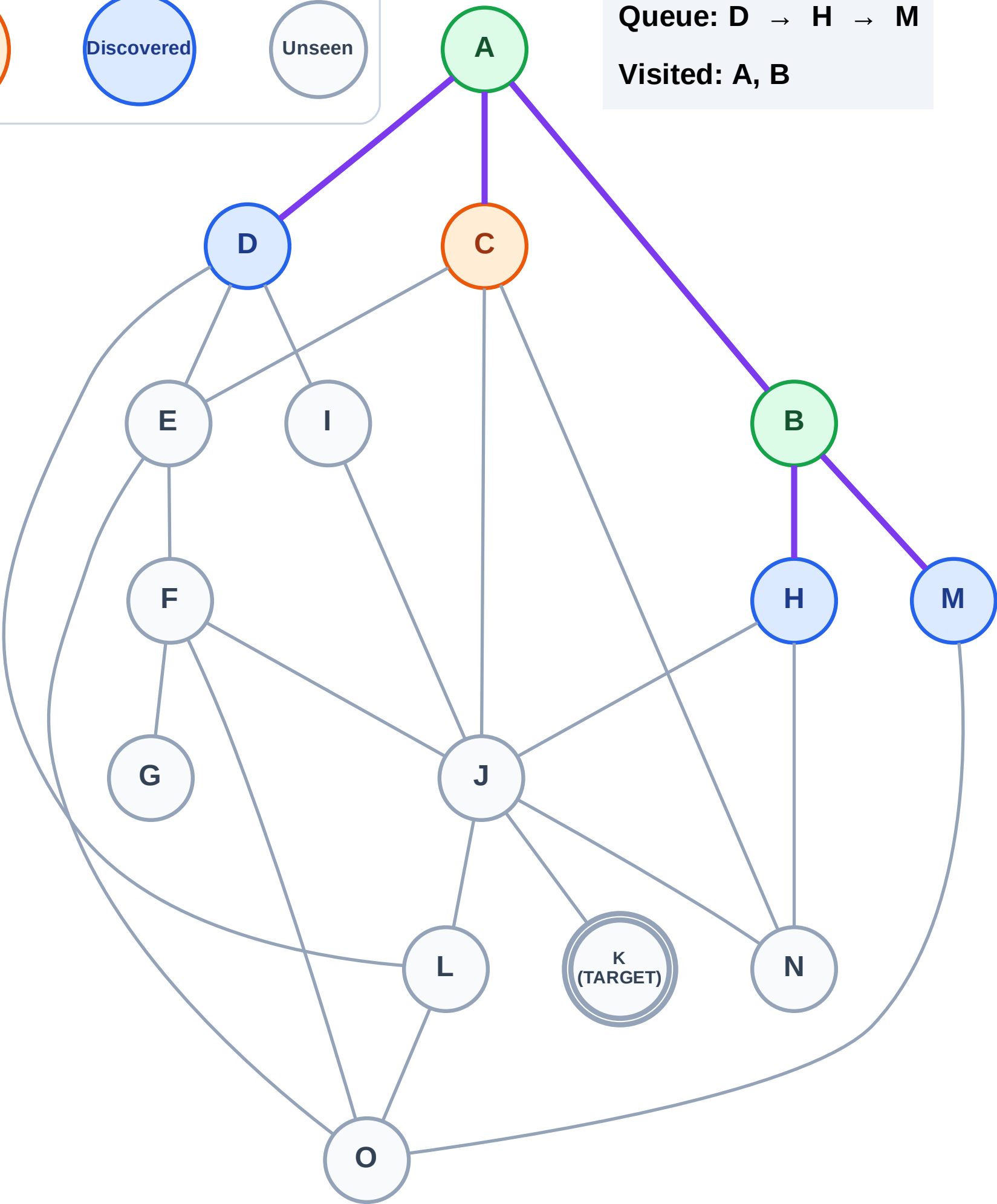
Processing

Discovered

Unseen

Queue: D → H → M

Visited: A, B

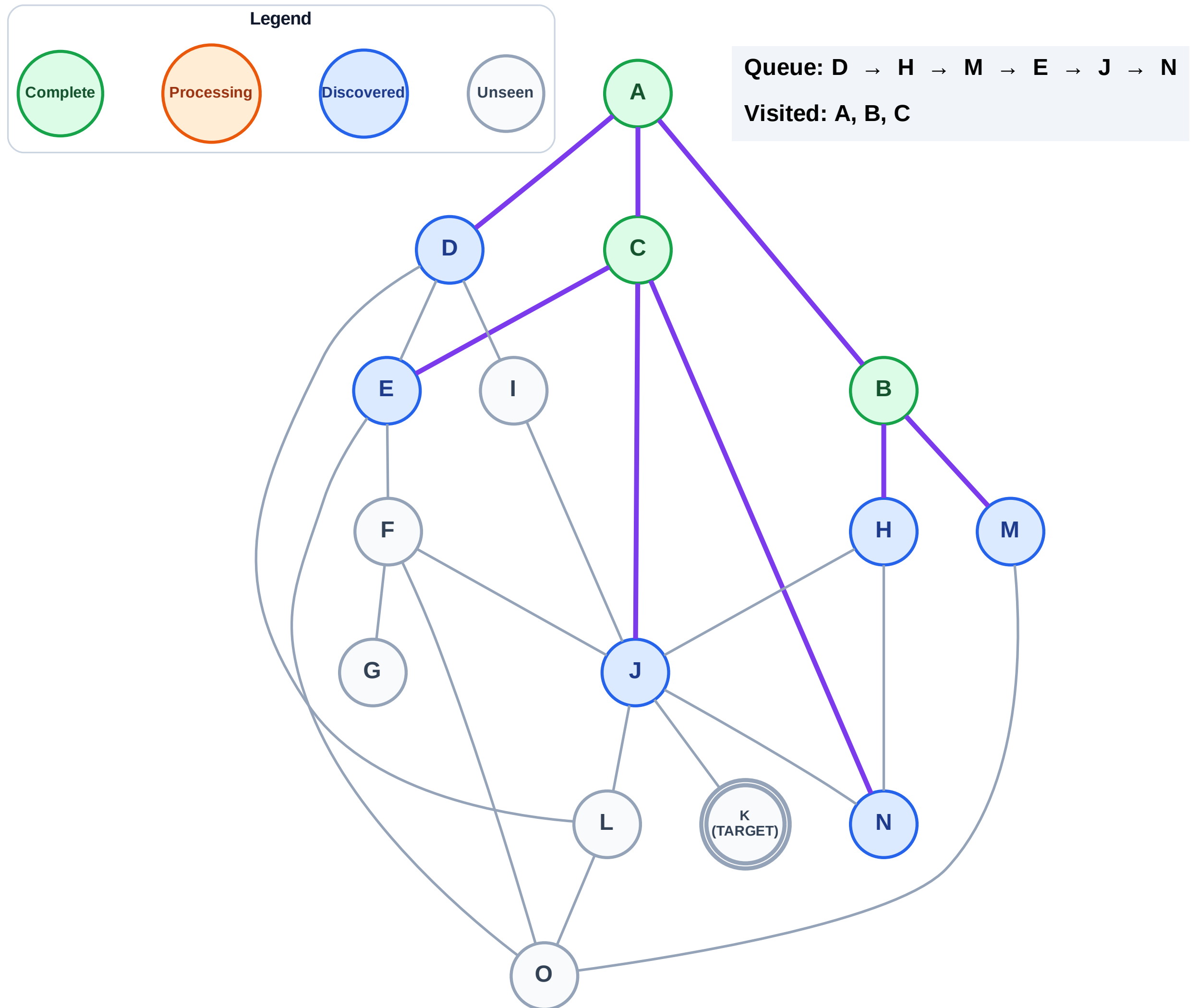


BFS Traversal

Step 7: Discover E, J, N from C

BFS Traversal

Step 7: Discover E, J, N from C



BFS Traversal

Step 8: Process node D

Legend

Complete

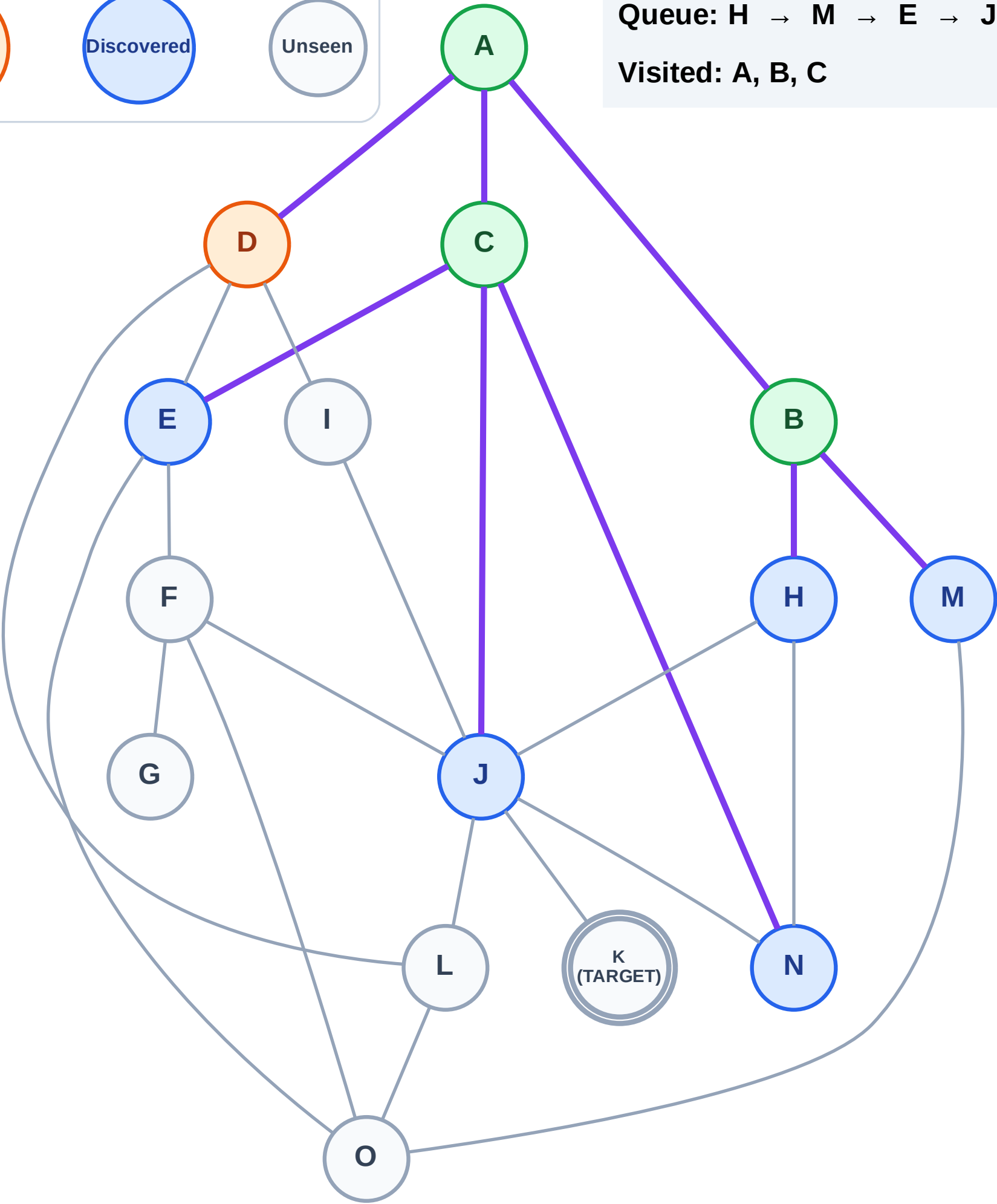
Processing

Discovered

Unseen

Queue: H → M → E → J → N

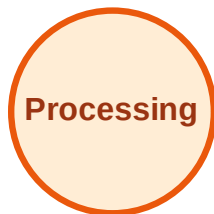
Visited: A, B, C



BFS Traversal

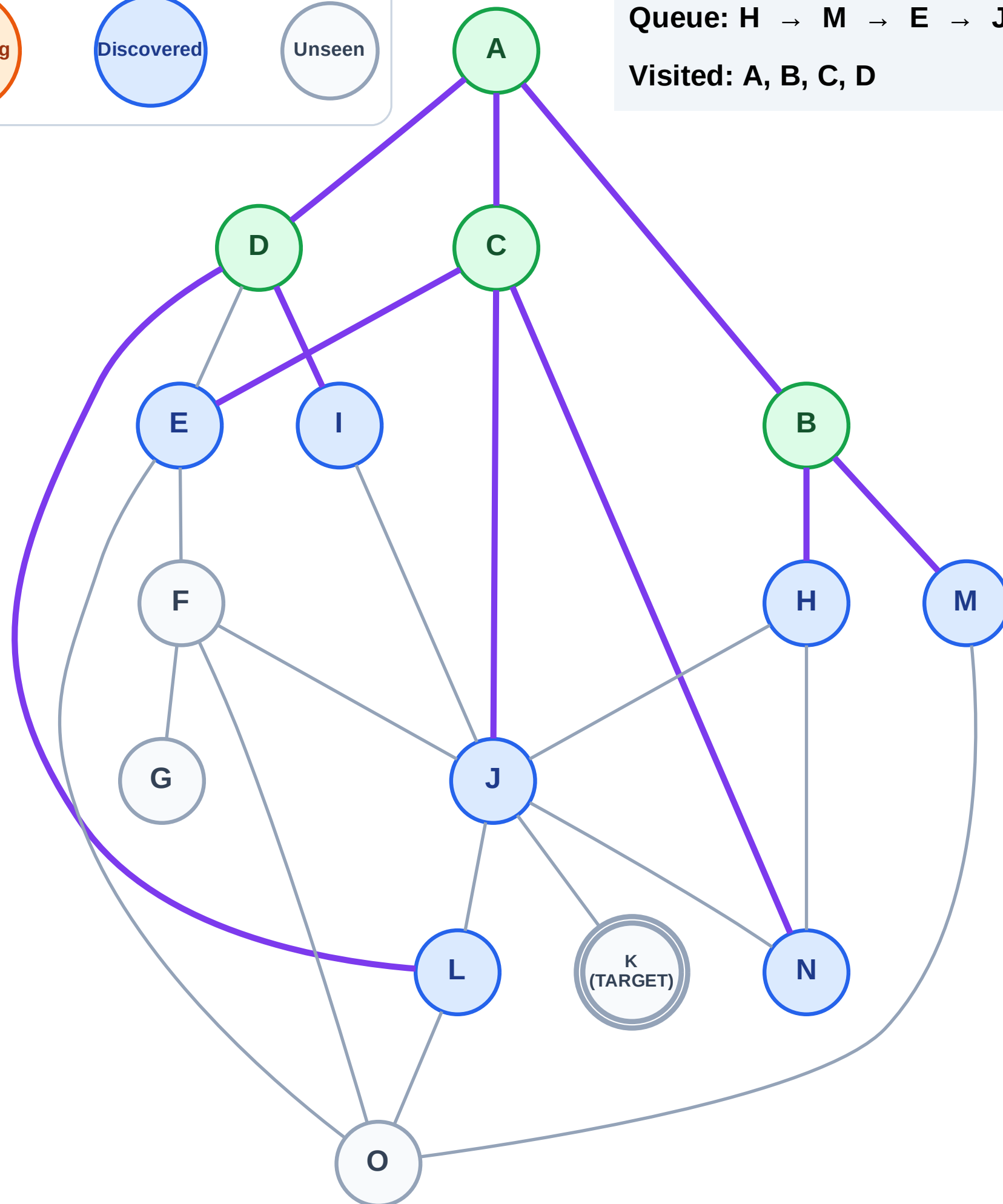
Step 9: Discover I, L from D

Legend



Queue: H → M → E → J → N → I → L

Visited: A, B, C, D



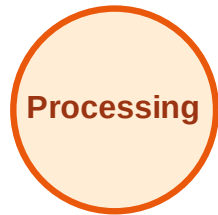
BFS Traversal

Step 10: Process node H

Legend



Complete



Processing



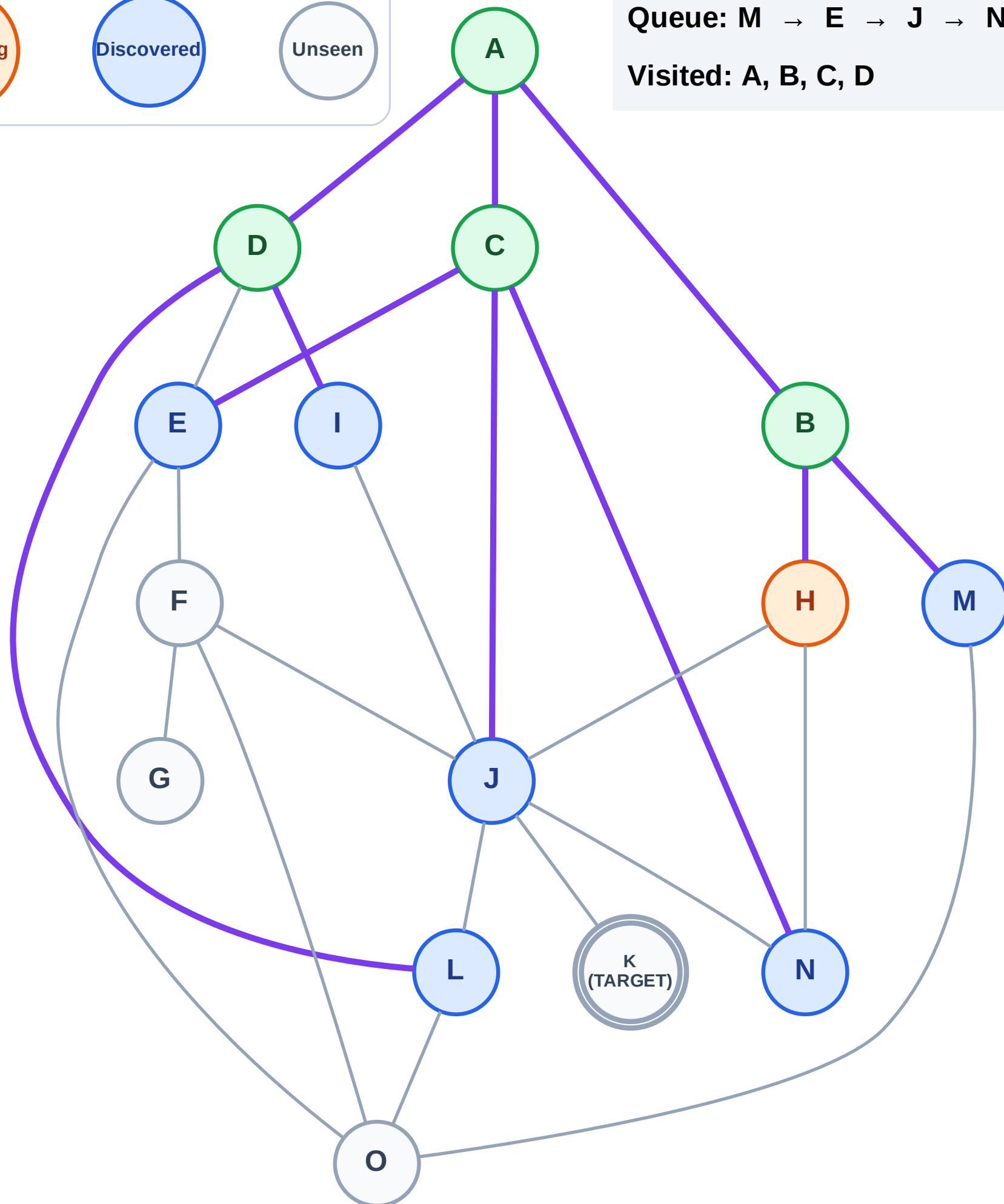
Discovered



Unseen

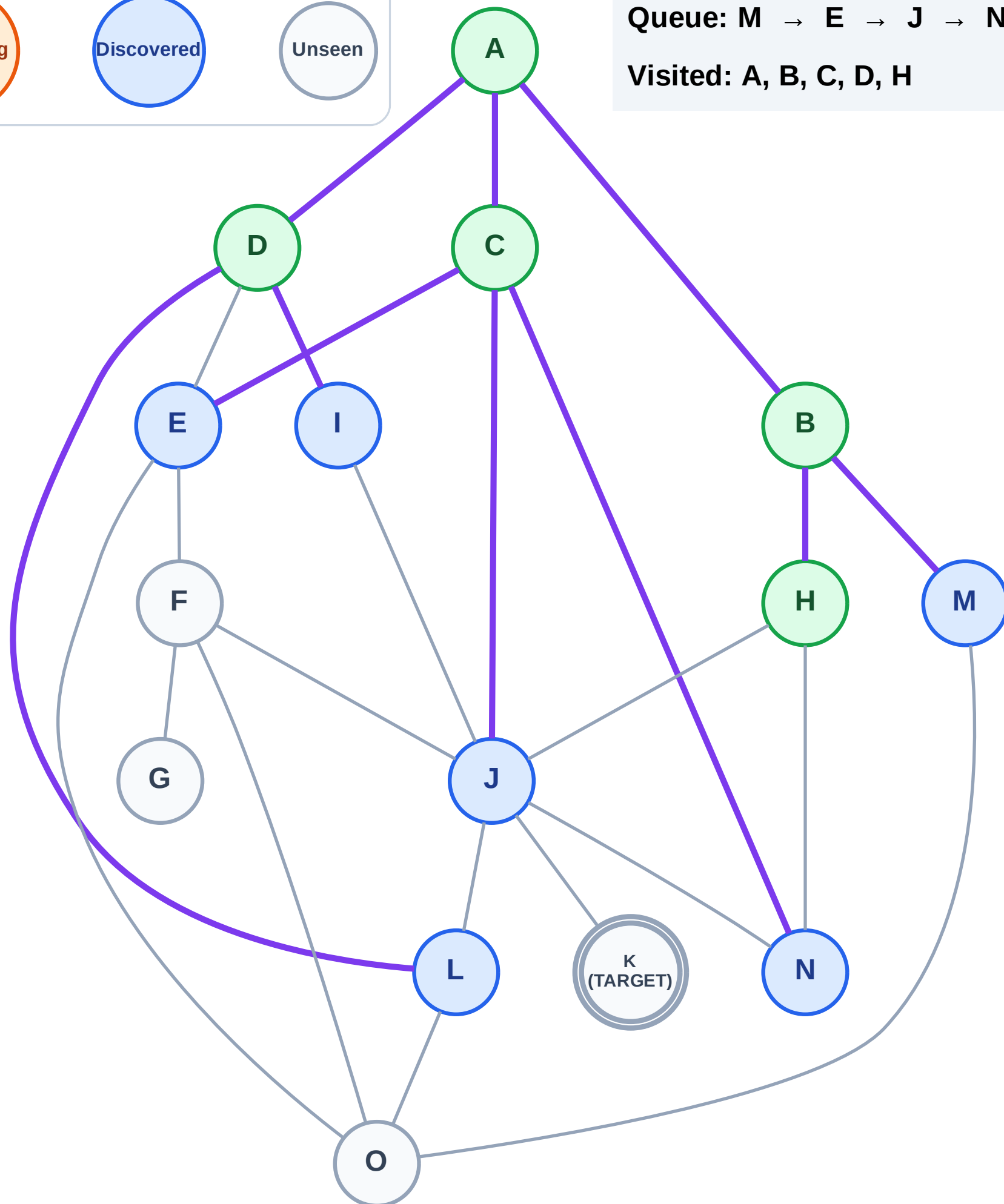
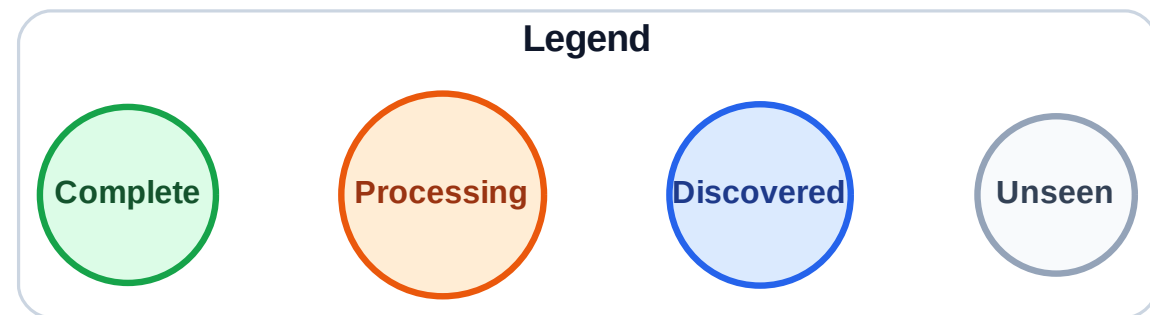
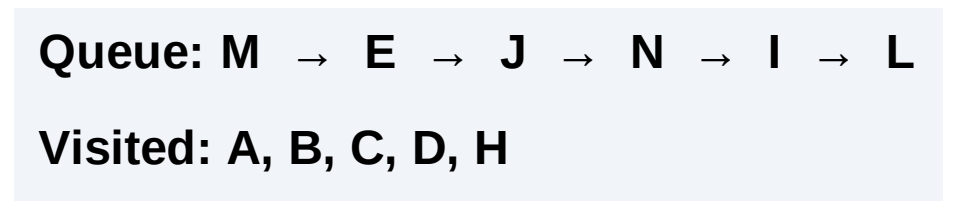
Queue: M → E → J → N → I → L

Visited: A, B, C, D



Step 11: No new neighbors from H

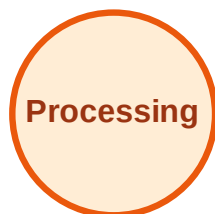
Step 11: No new neighbors from H



BFS Traversal

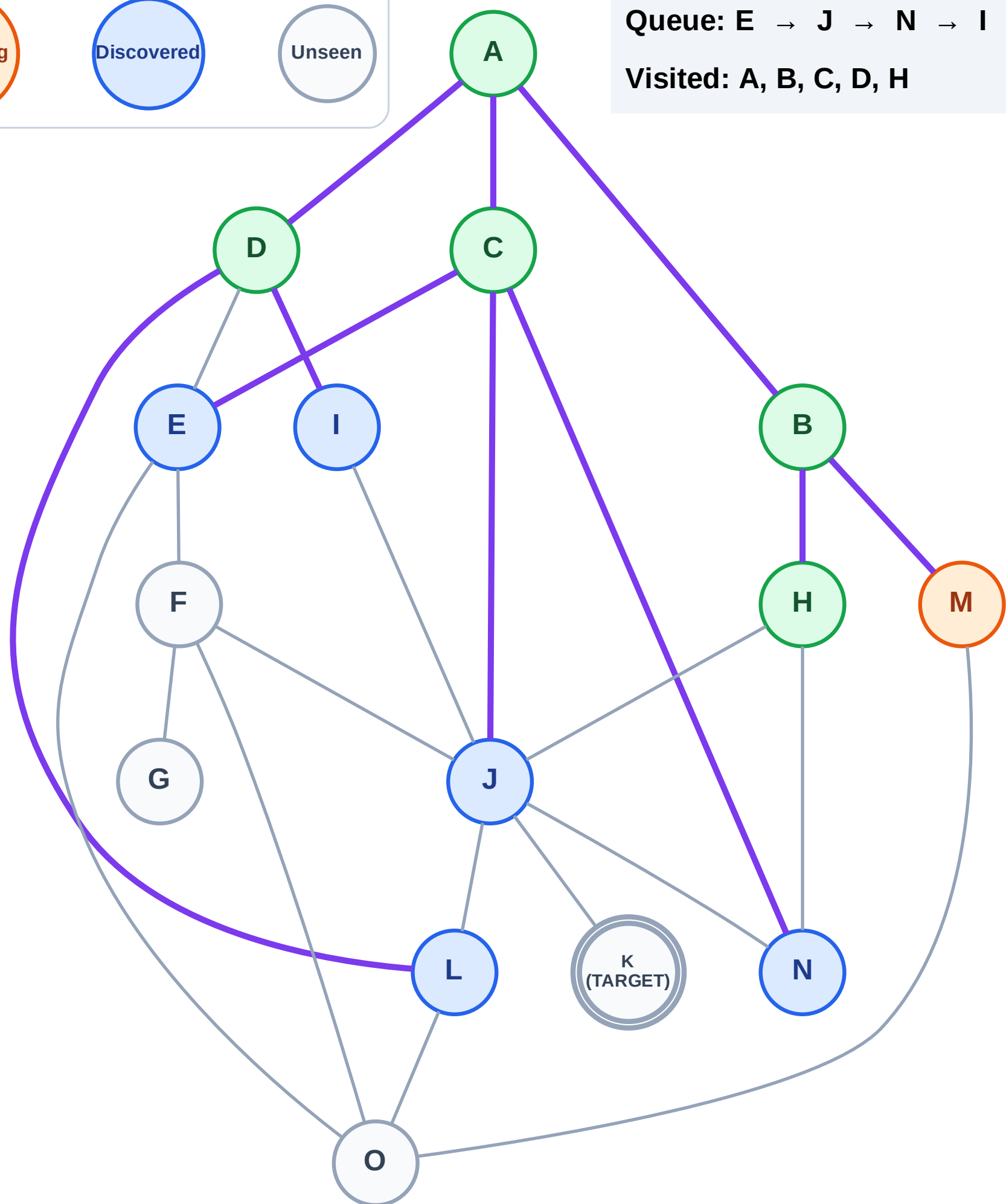
Step 12: Process node M

Legend



Queue: E → J → N → I → L

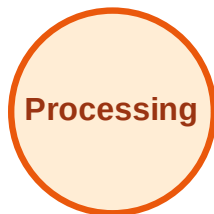
Visited: A, B, C, D, H



BFS Traversal

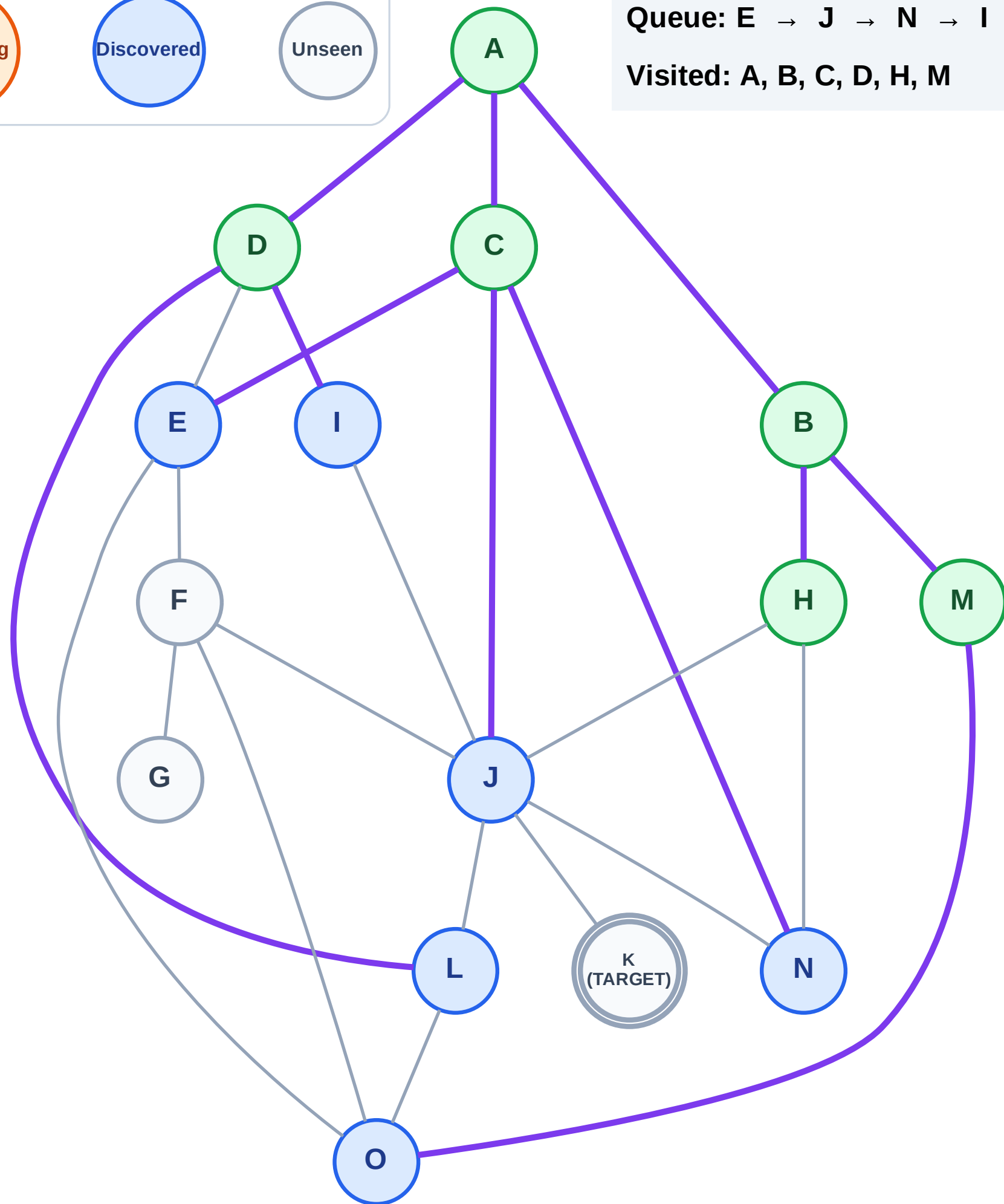
Step 13: Discover O from M

Legend



Queue: E → J → N → I → L → O

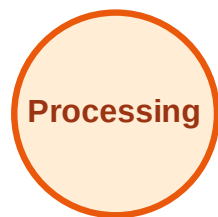
Visited: A, B, C, D, H, M



BFS Traversal

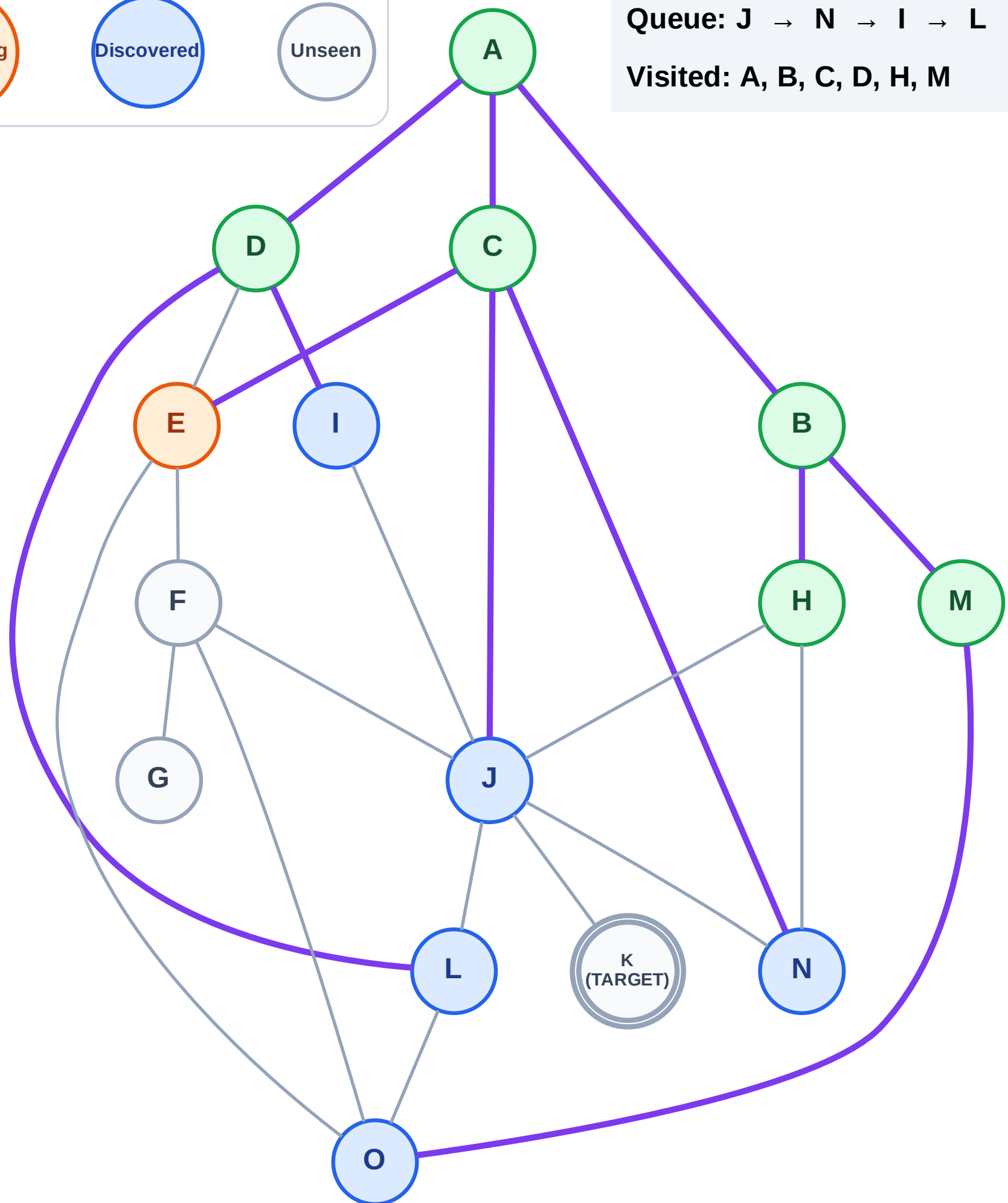
Step 14: Process node E

Legend



Queue: J → N → I → L → O

Visited: A, B, C, D, H, M



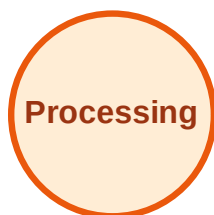
BFS Traversal

Step 15: Discover F from E

Legend



Complete



Processing



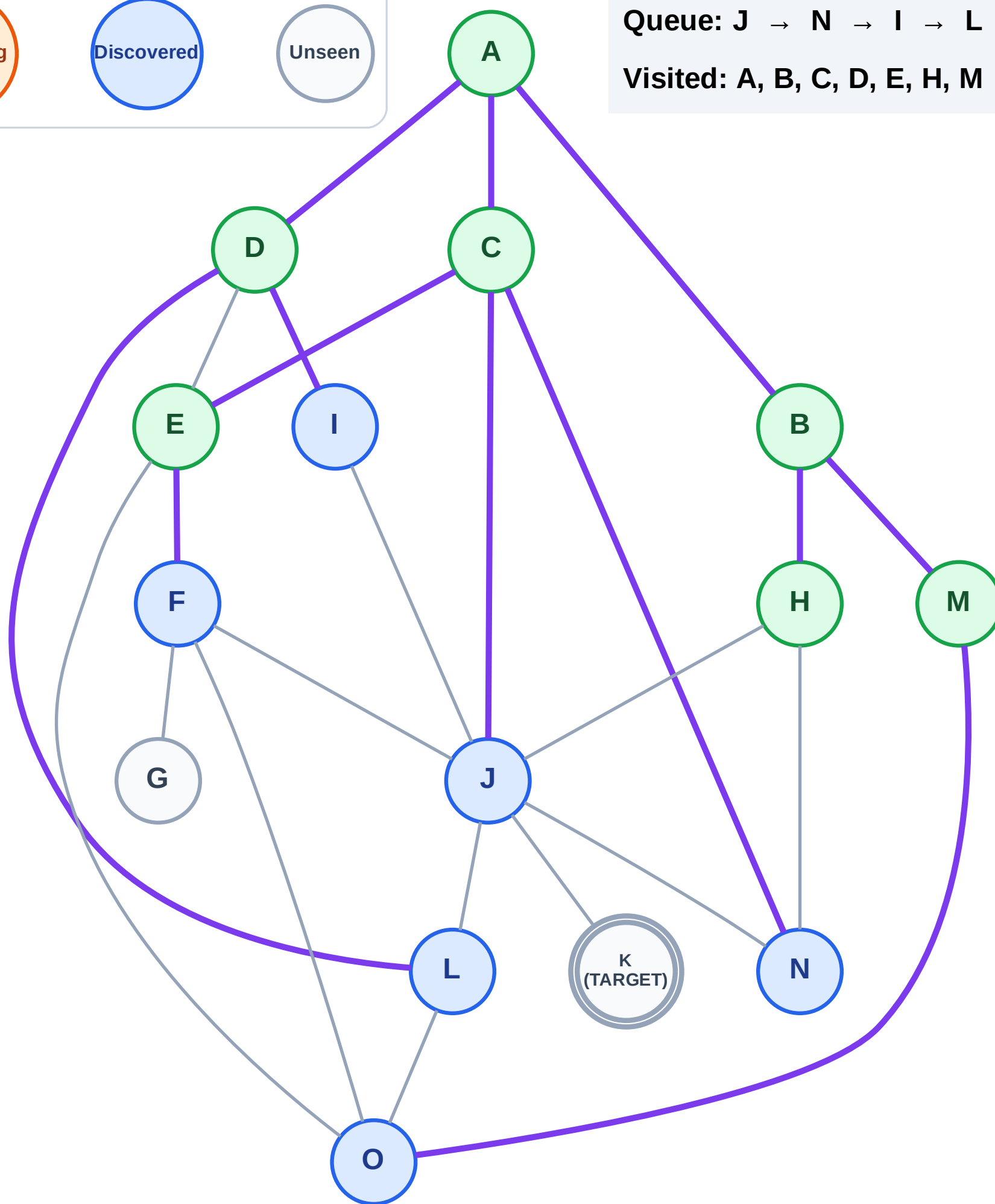
Discovered



Unseen

Queue: J → N → I → L → O → F

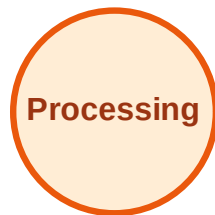
Visited: A, B, C, D, E, H, M



BFS Traversal

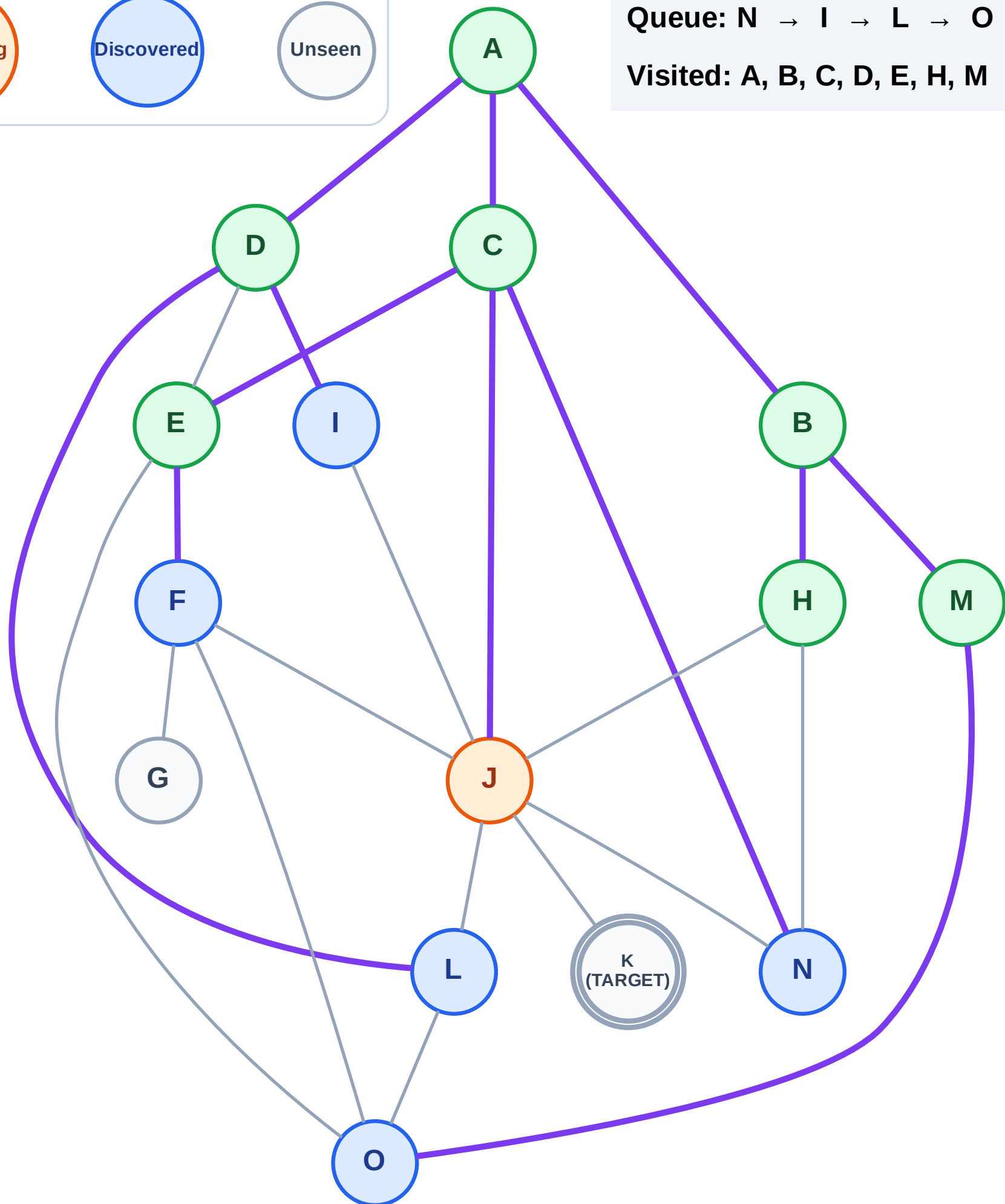
Step 16: Process node J

Legend



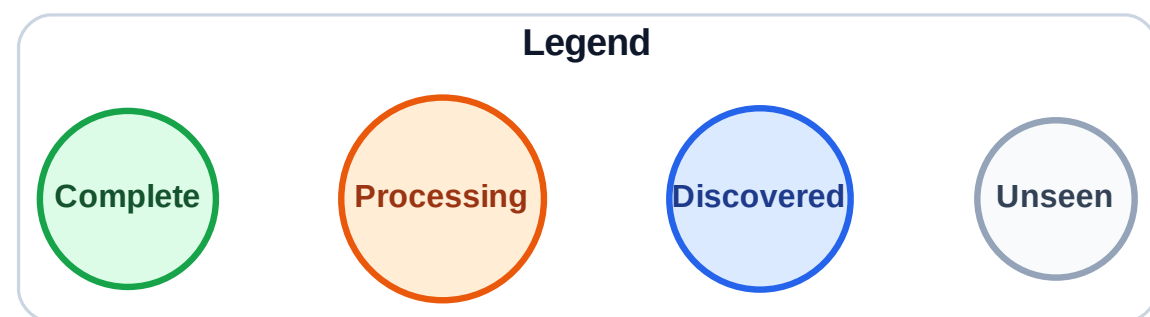
Queue: N → I → L → O → F

Visited: A, B, C, D, E, H, M

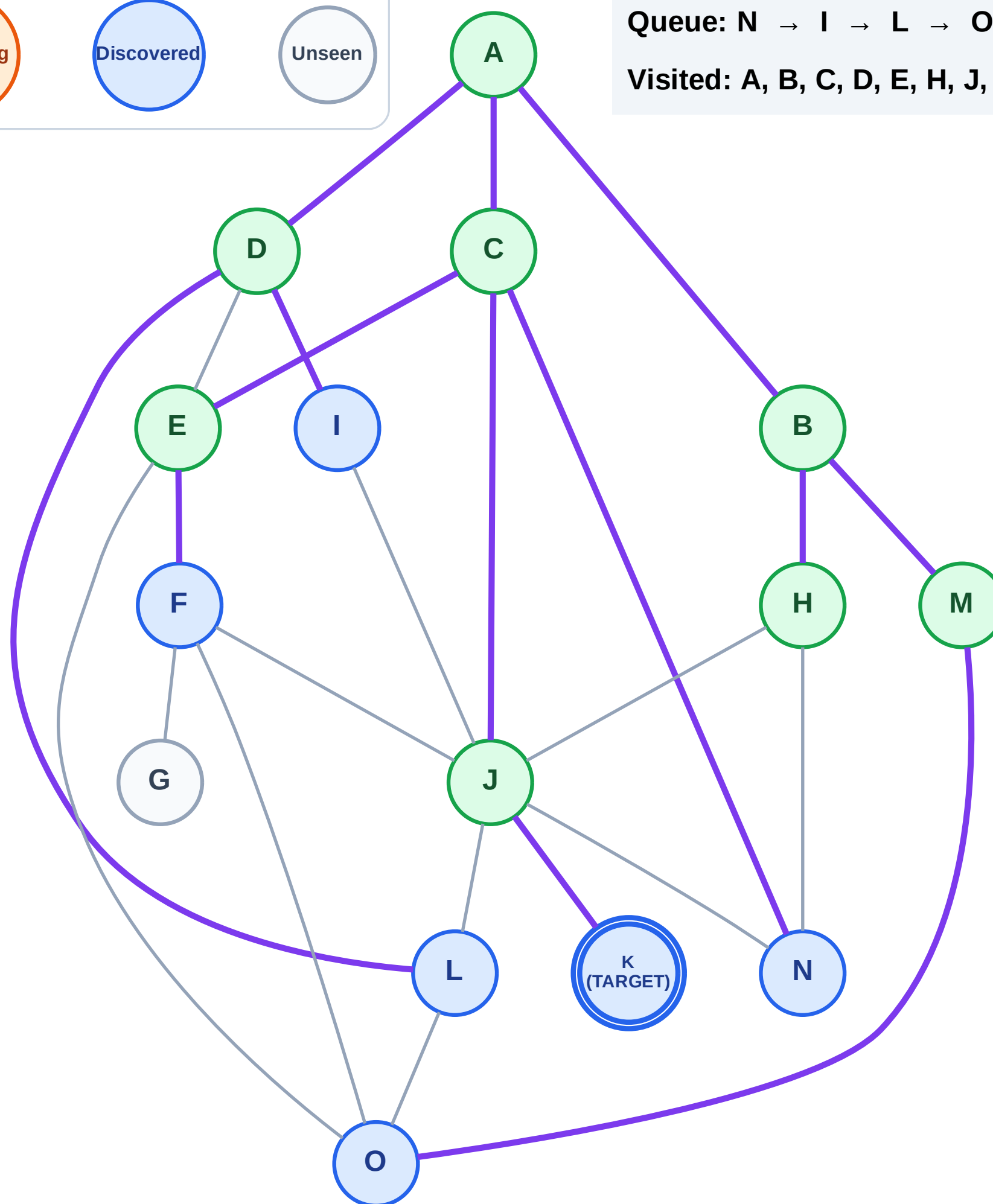


BFS Traversal

Step 17: Discover K from J



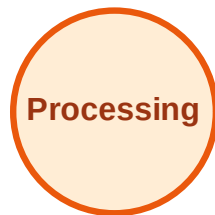
Queue: N → I → L → O → F → K
Visited: A, B, C, D, E, H, J, M



BFS Traversal

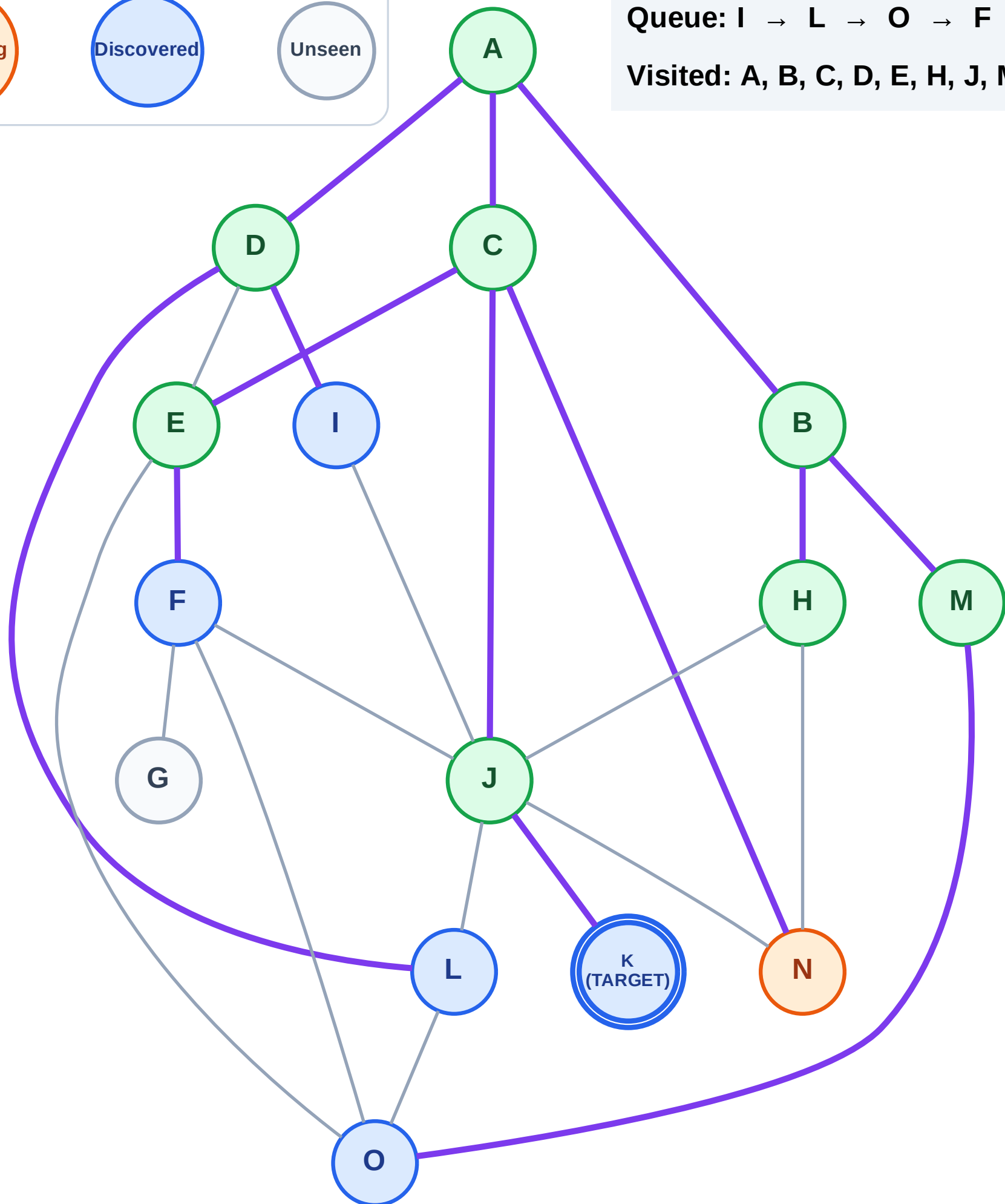
Step 18: Process node N

Legend



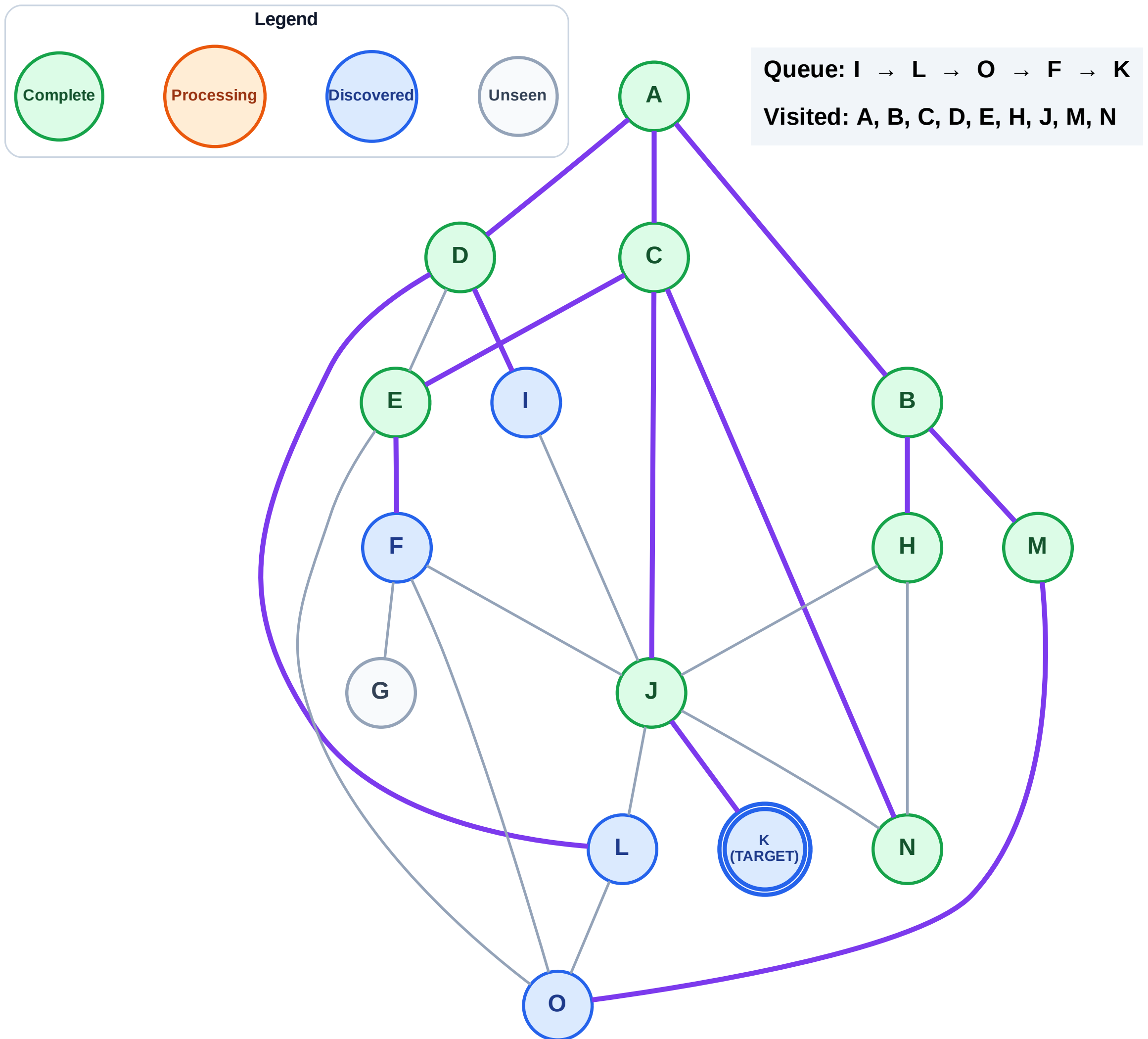
Queue: I → L → O → F → K

Visited: A, B, C, D, E, H, J, M



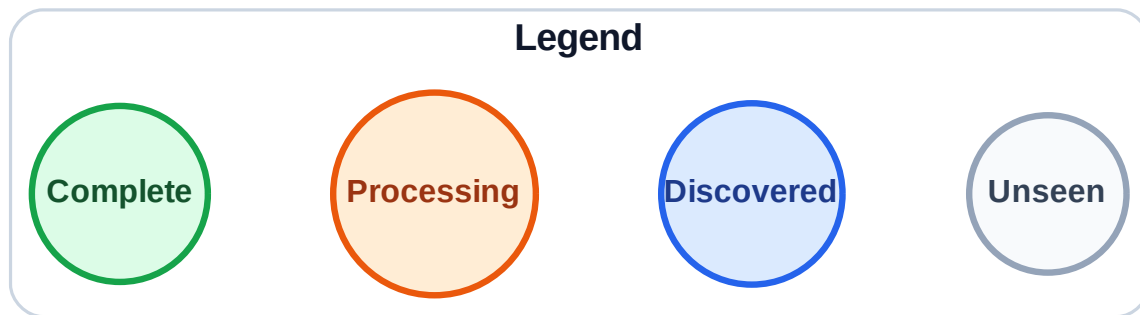
Step 19: No new neighbors from N

Step 19: No new neighbors from N



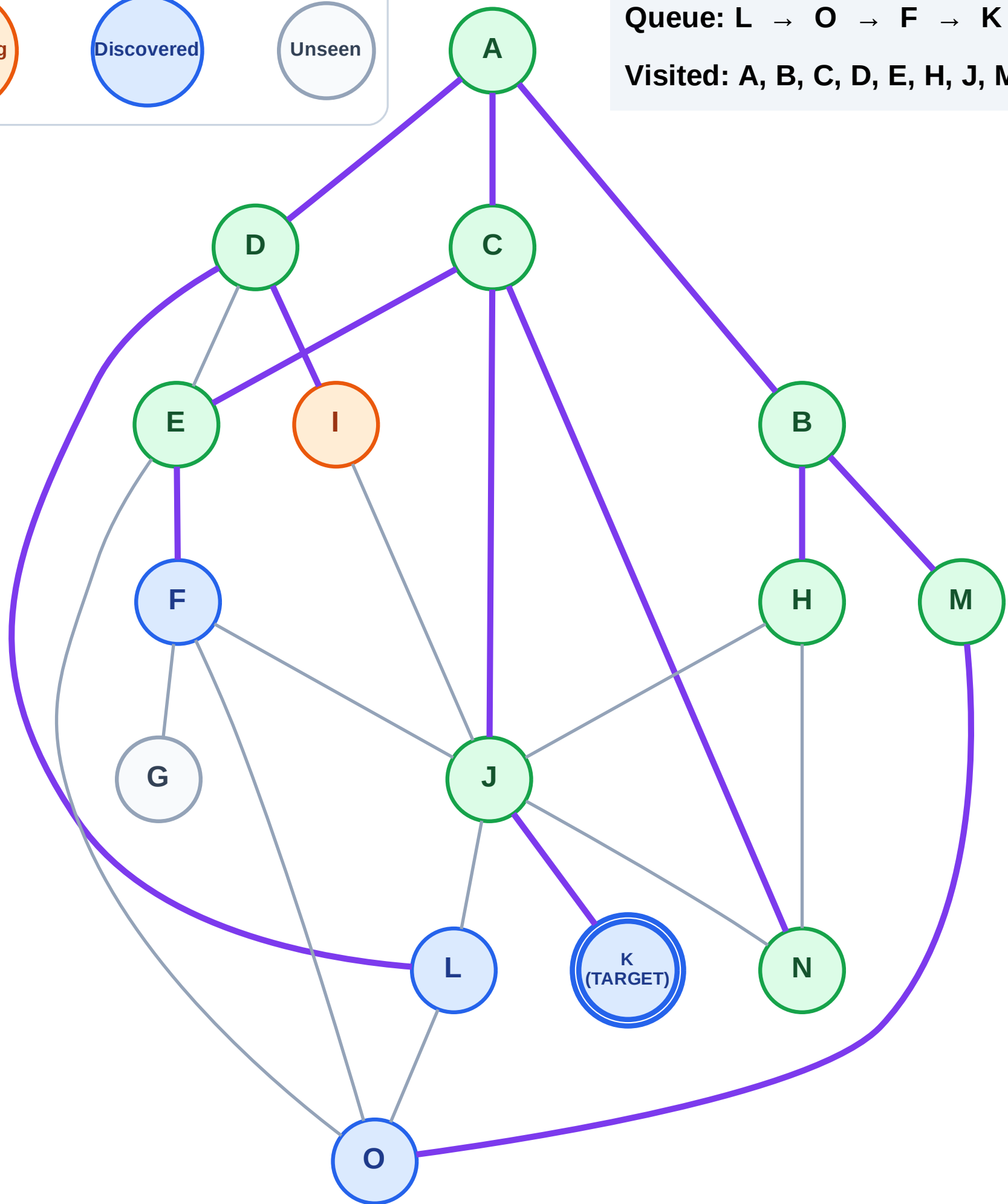
Step 20: Process node I

Step 20: Process node I



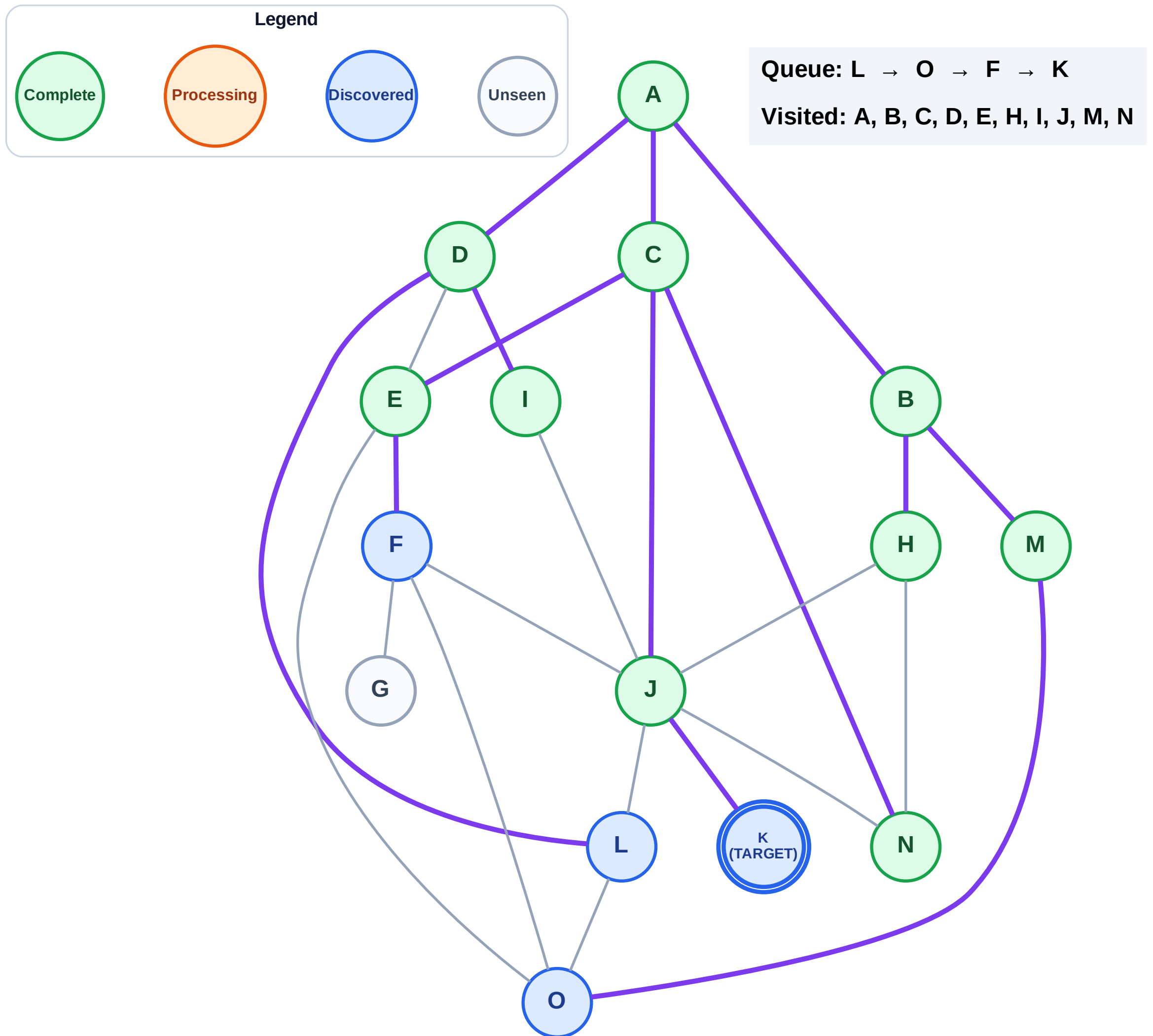
Queue: L → O → F → K

Visited: A, B, C, D, E, H, J, M, N



Step 21: No new neighbors from I

Step 21: No new neighbors from I



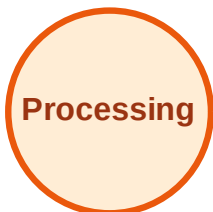
BFS Traversal

Step 22: Process node L

Legend



Complete



Processing



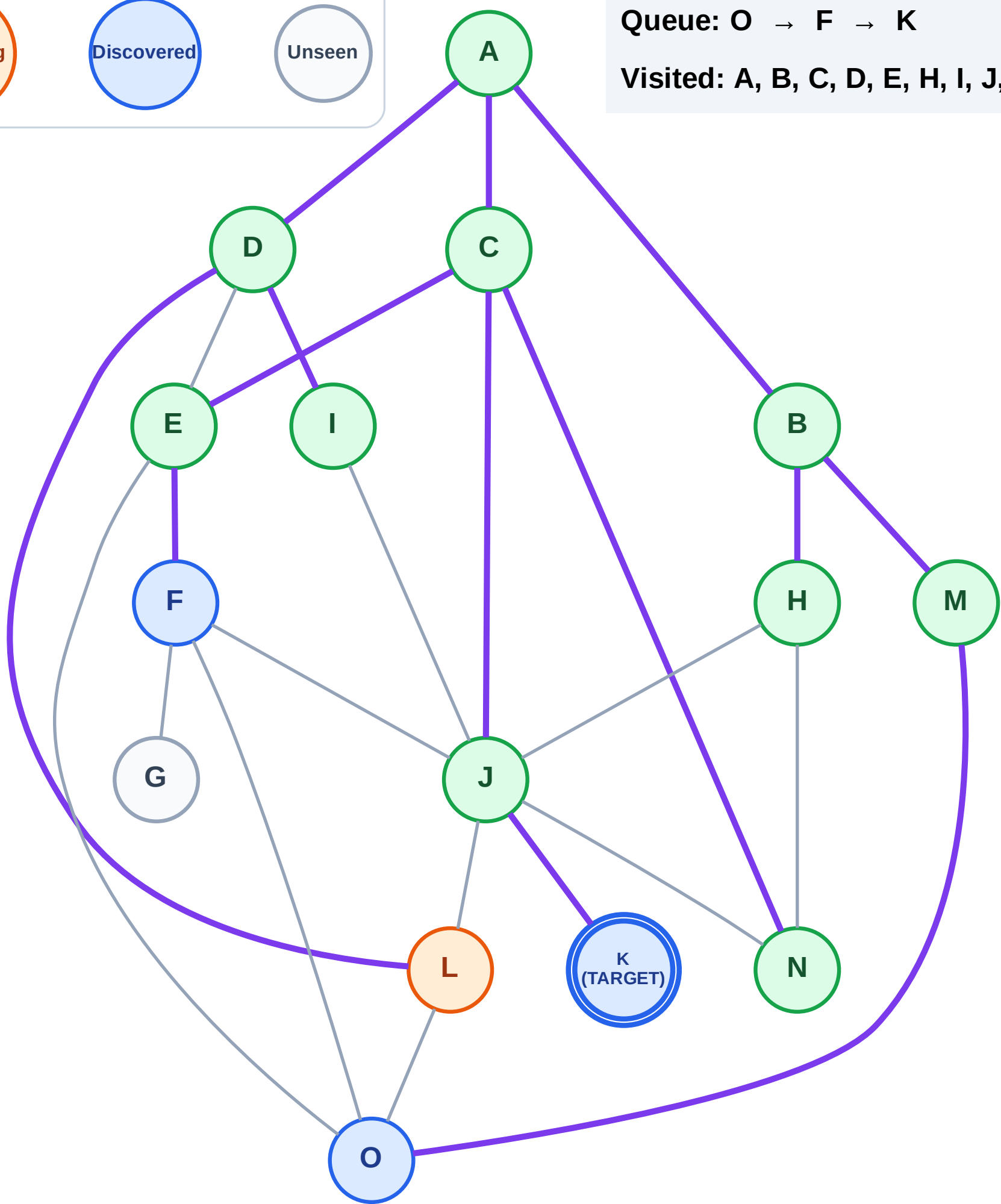
Discovered



Unseen

Queue: O → F → K

Visited: A, B, C, D, E, H, I, J, M, N



BFS Traversal

Step 23: No new neighbors from L

Legend

Complete

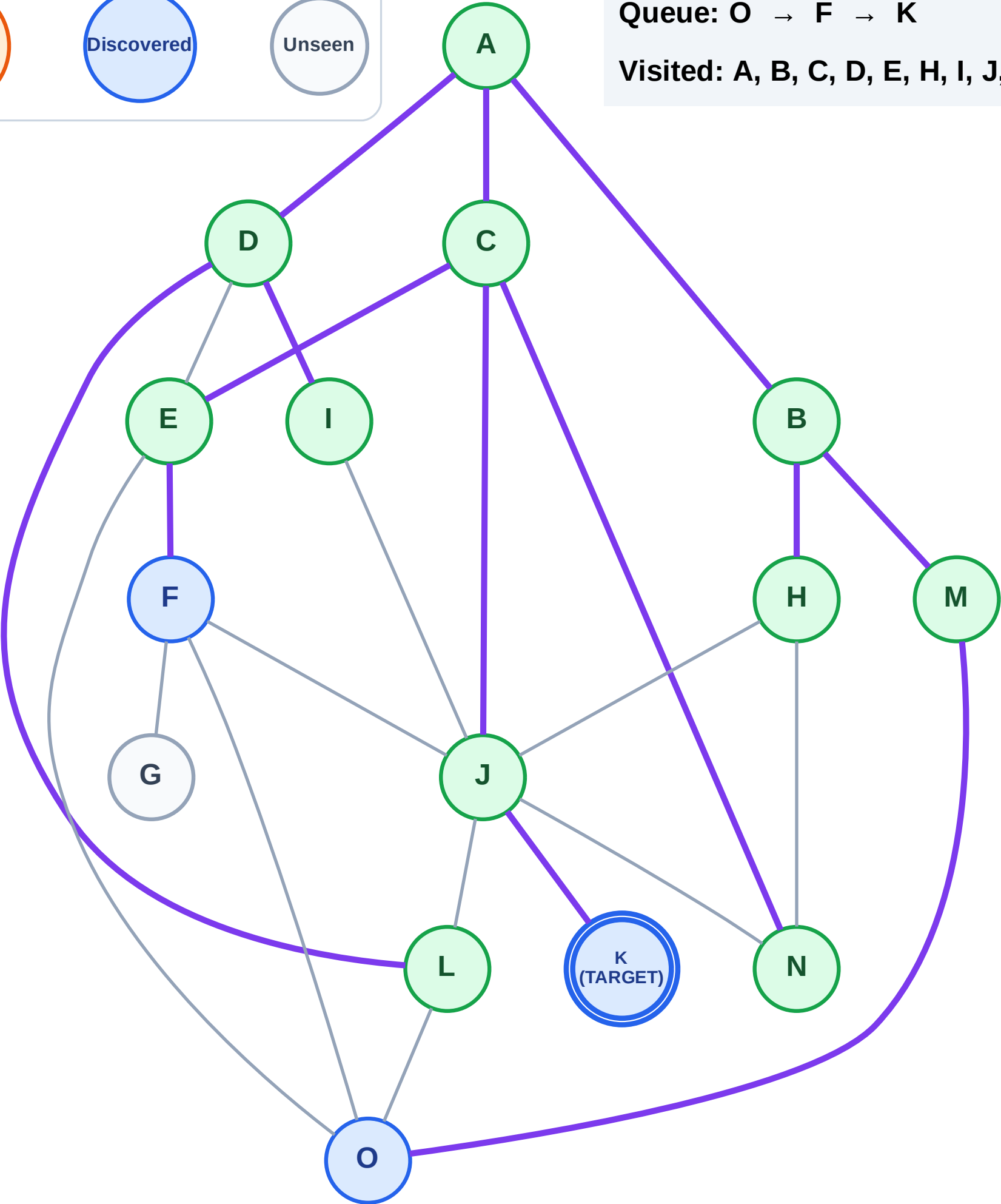
Processing

Discovered

Unseen

Queue: O → F → K

Visited: A, B, C, D, E, H, I, J, L, M, N



BFS Traversal

Step 24: Process node O

Legend

Complete

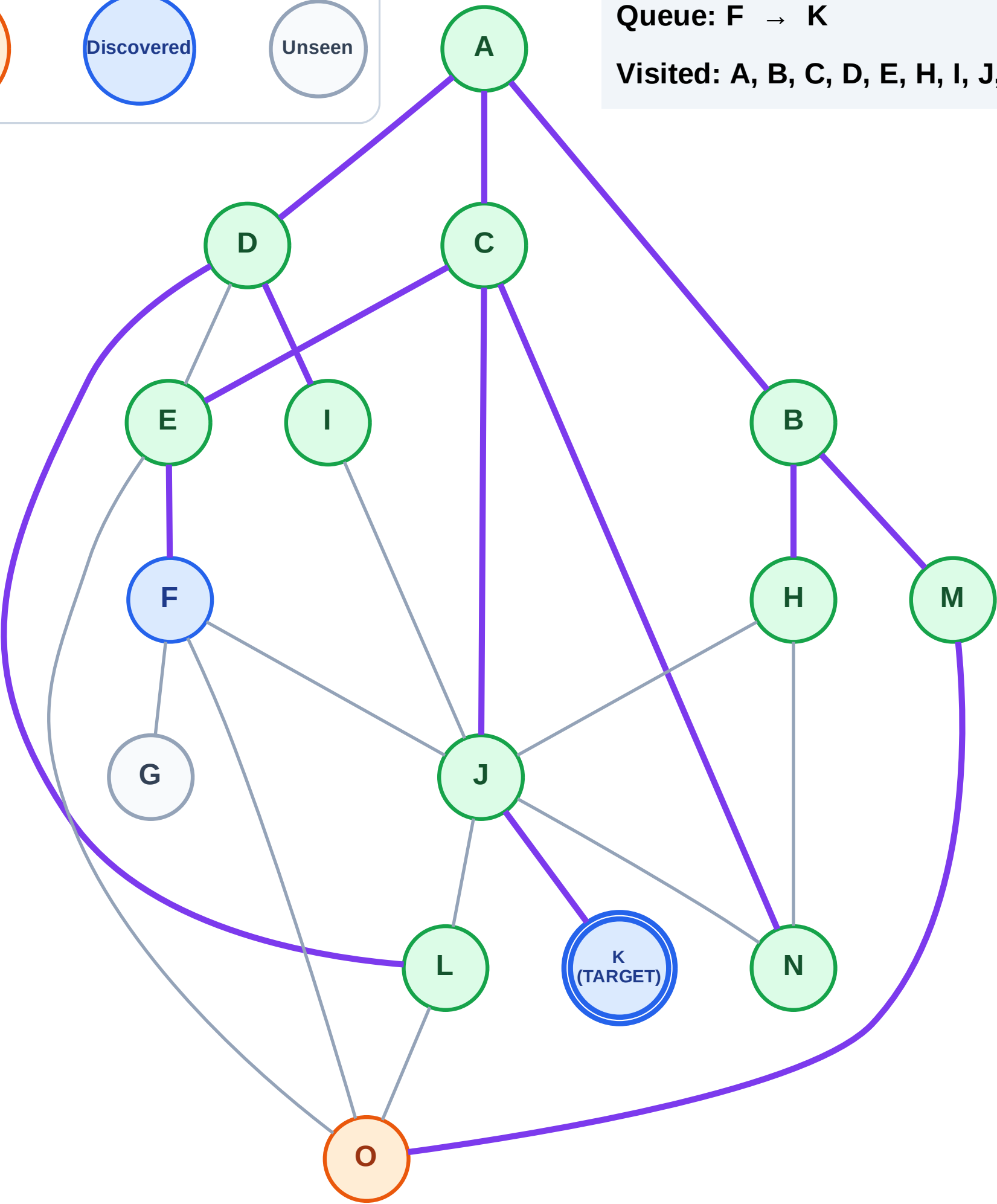
Processing

Discovered

Unseen

Queue: F → K

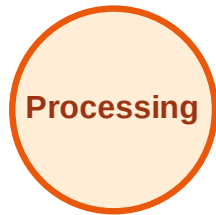
Visited: A, B, C, D, E, H, I, J, L, M, N



BFS Traversal

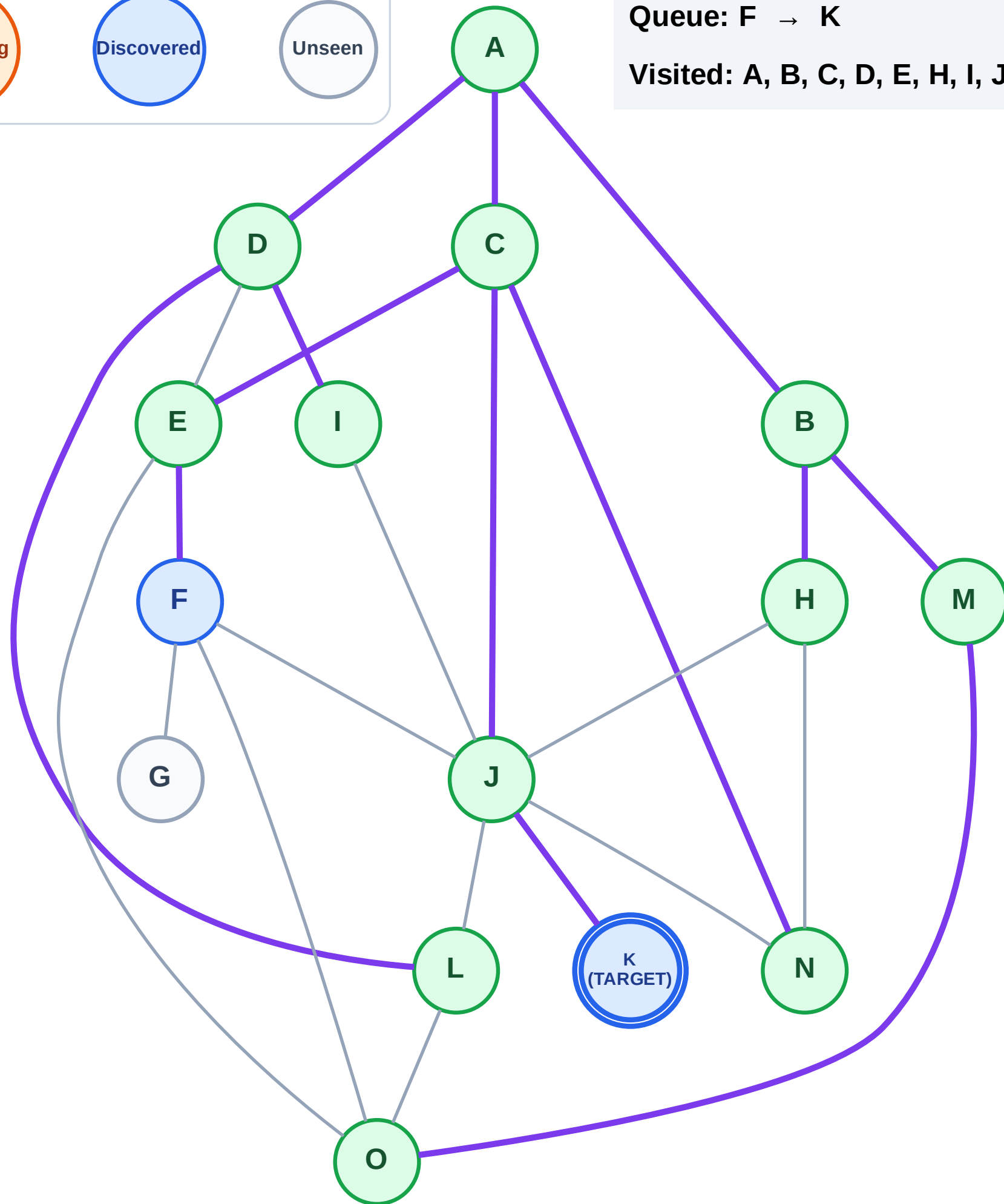
Step 25: No new neighbors from O

Legend



Queue: F → K

Visited: A, B, C, D, E, H, I, J, L, M, N, O



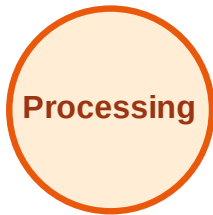
BFS Traversal

Step 26: Process node F

Legend



Complete



Processing



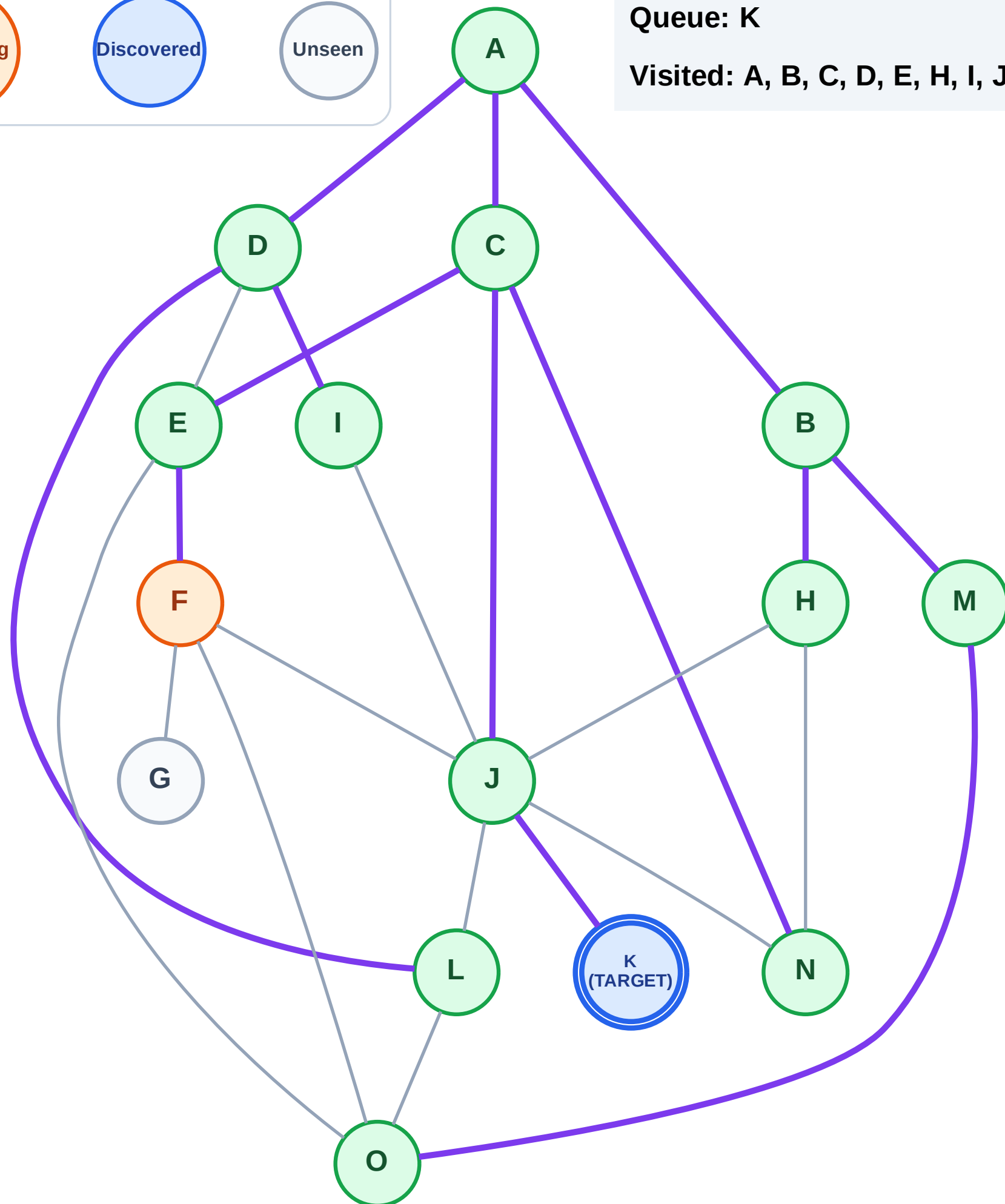
Discovered



Unseen

Queue: K

Visited: A, B, C, D, E, H, I, J, L, M, N, O



BFS Traversal

Step 27: Discover G from F

Legend

Complete

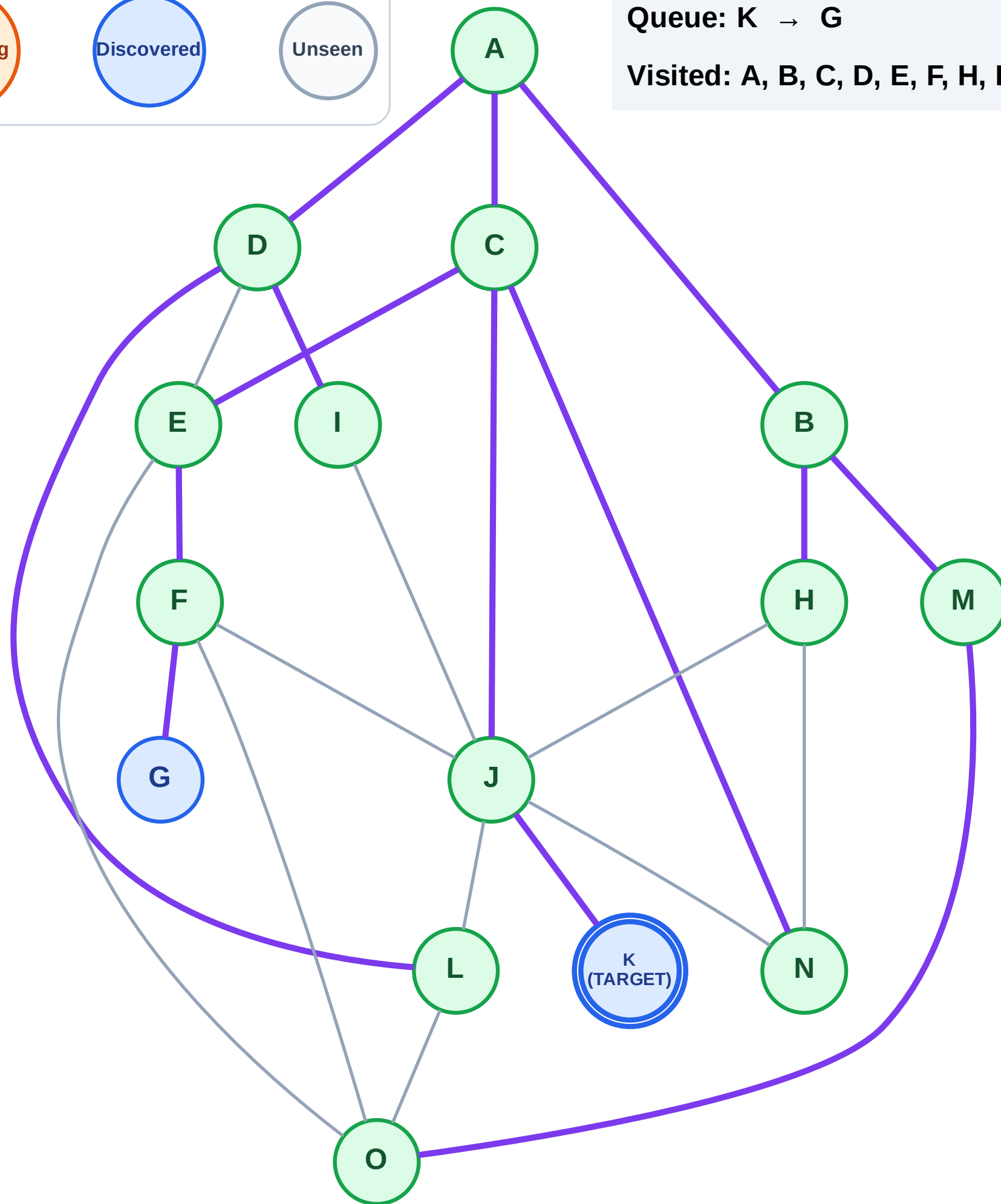
Processing

Discovered

Unseen

Queue: K → G

Visited: A, B, C, D, E, F, H, I, J, L, M, N, O



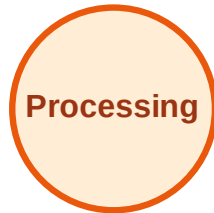
BFS Traversal

Step 28: Process node K

Legend



Complete



Processing



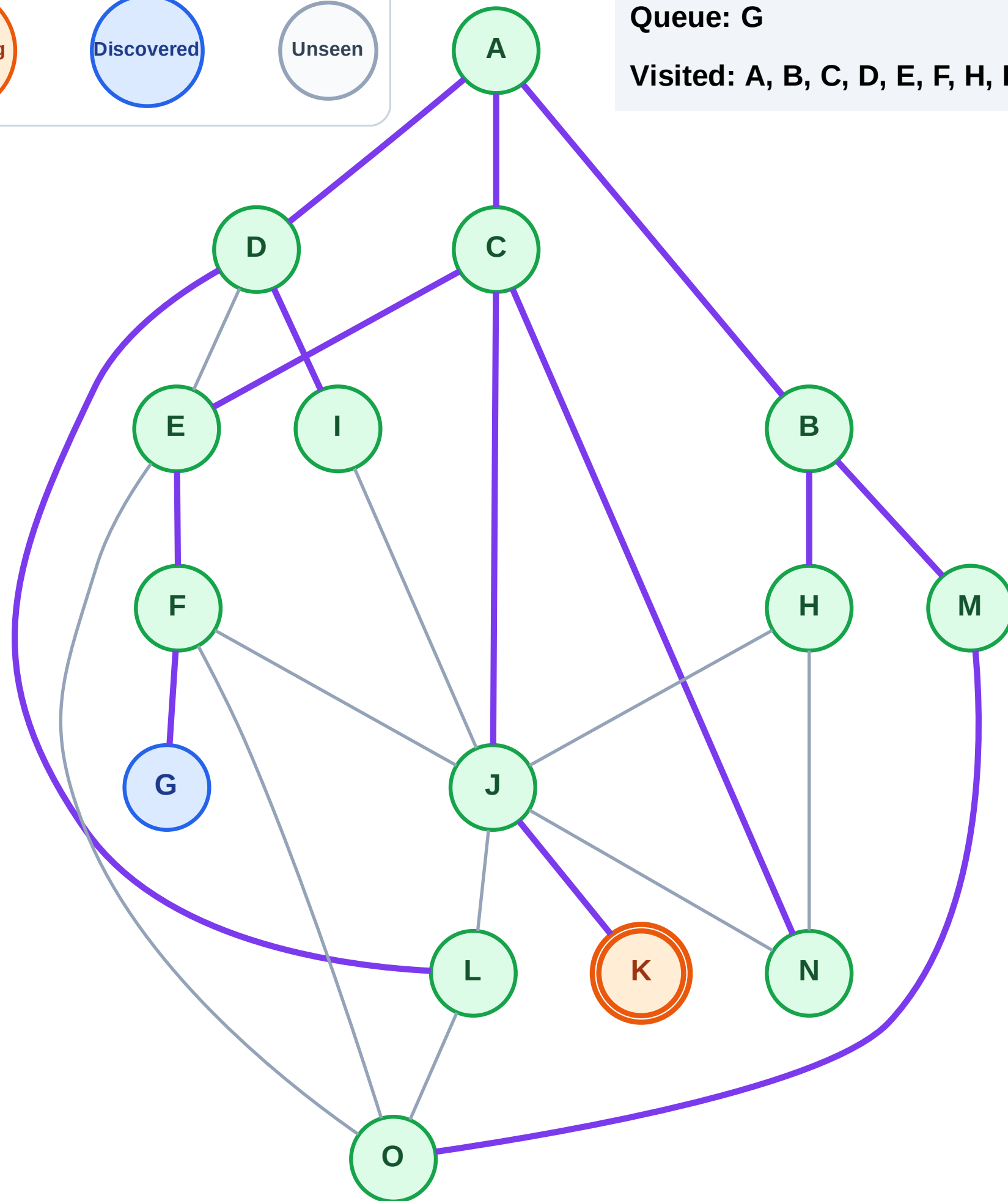
Discovered



Unseen

Queue: G

Visited: A, B, C, D, E, F, H, I, J, L, M, N, O



BFS Traversal

Step 29: TARGET NODE K REACHED! Path (3 hops): A → C → J → K

Legend



Complete



Processing



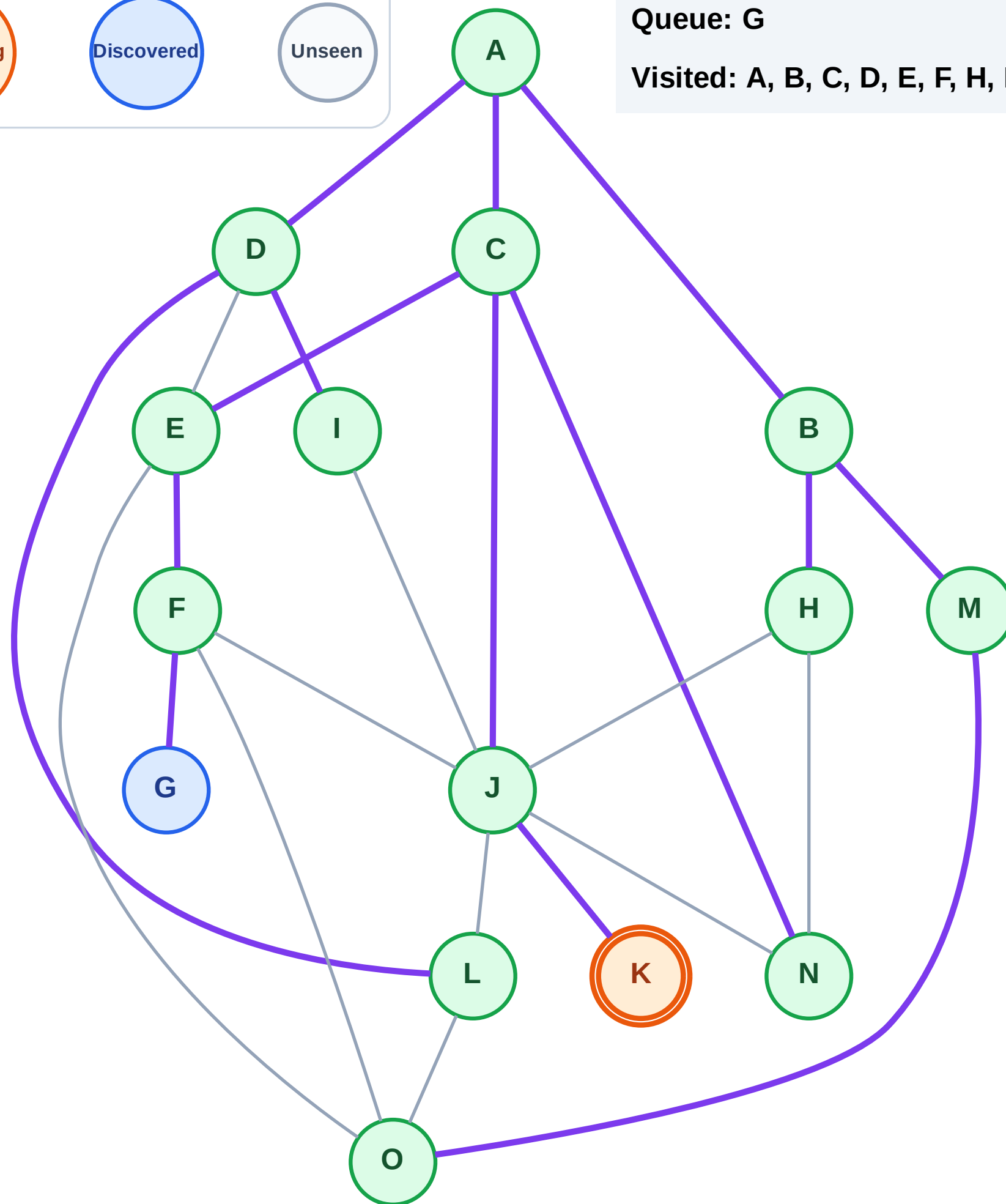
Discovered



Unseen

Queue: G

Visited: A, B, C, D, E, F, H, I, J, K, L, M, N, O



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

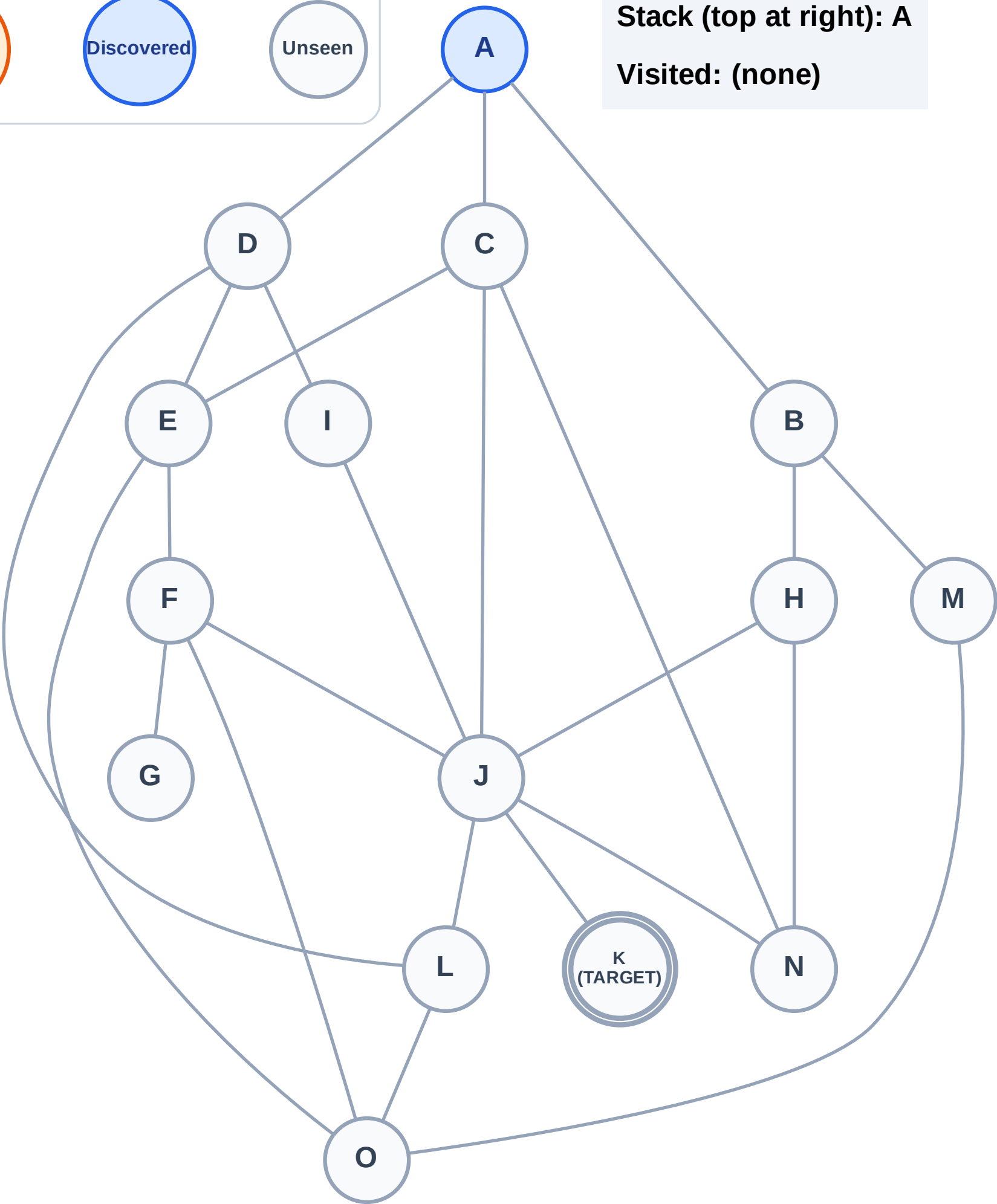
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

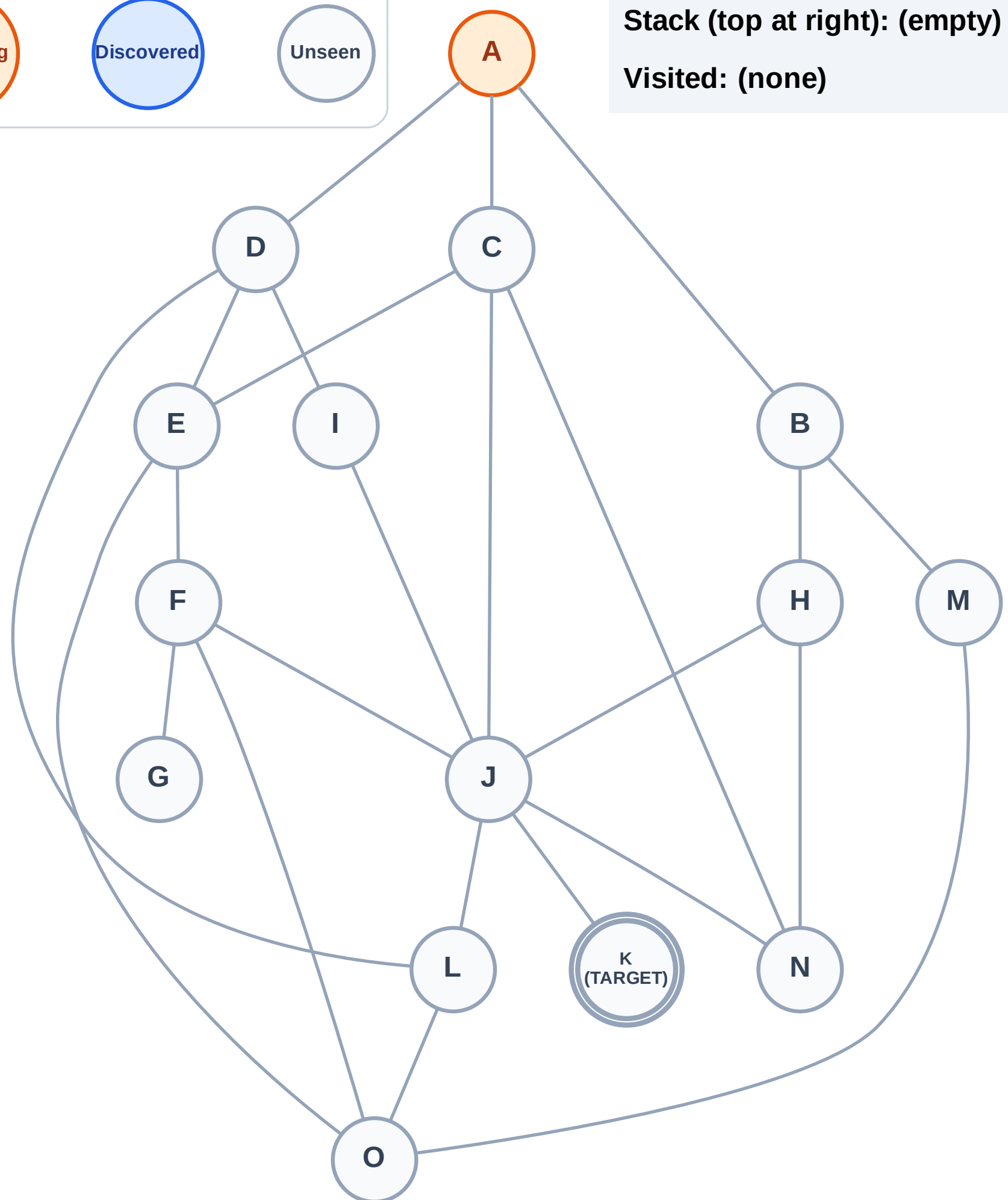
Processing

Discovered

Unseen

Stack (top at right): (empty)

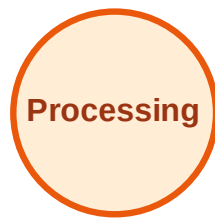
Visited: (none)



DFS Traversal

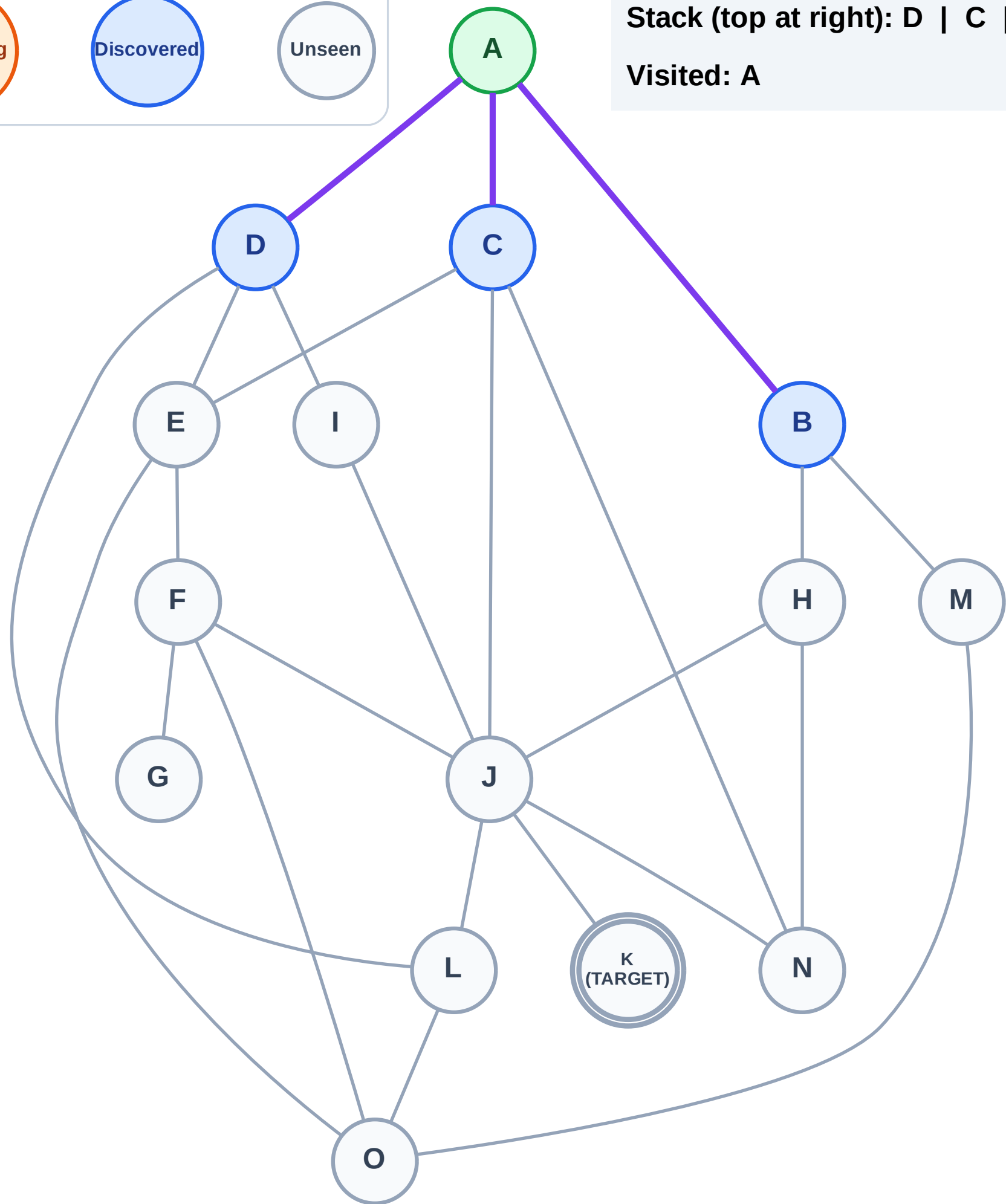
Step 3: Discover D, C, B from A

Legend



Stack (top at right): D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

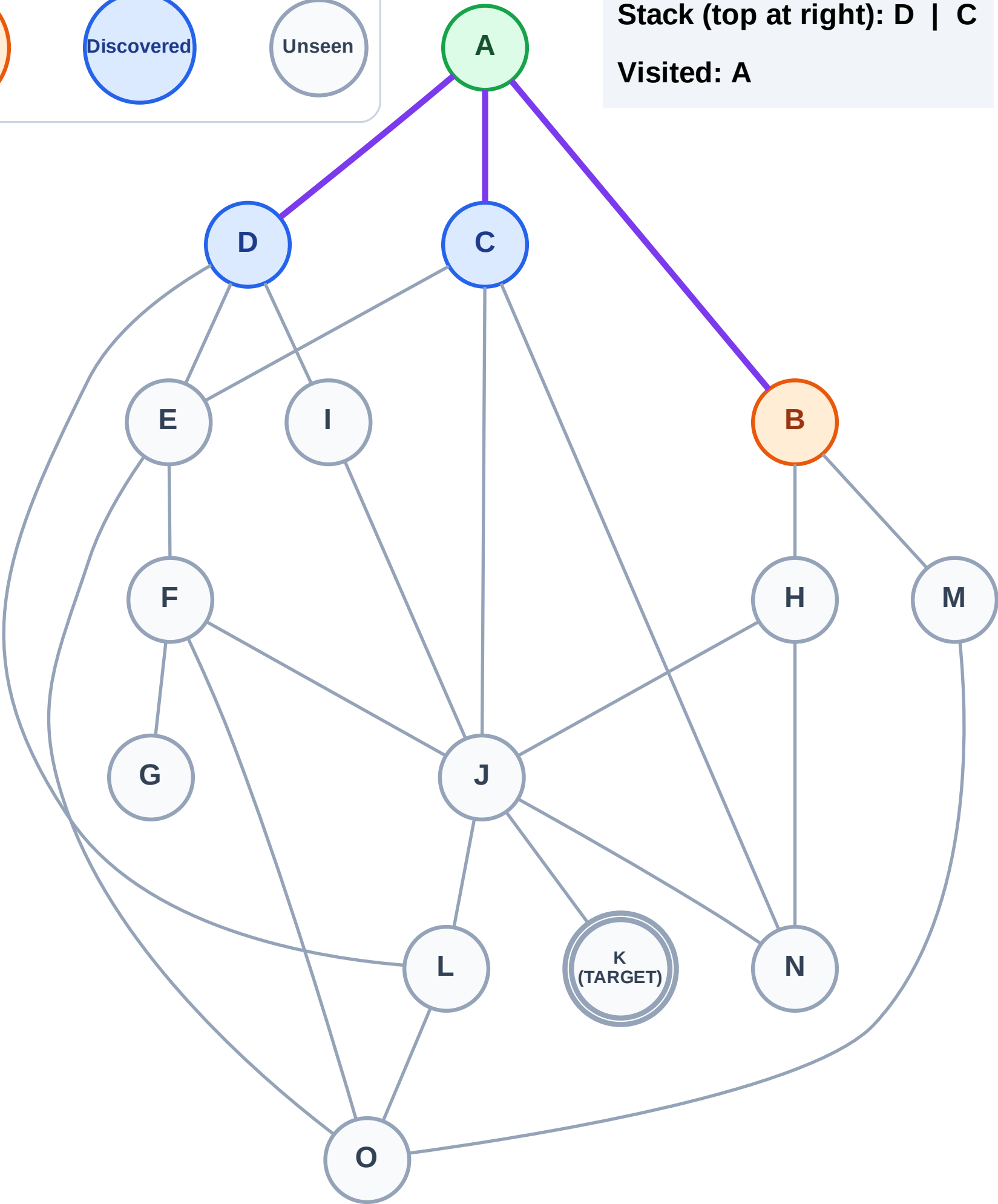
Processing

Discovered

Unseen

Stack (top at right): D | C

Visited: A



DFS Traversal

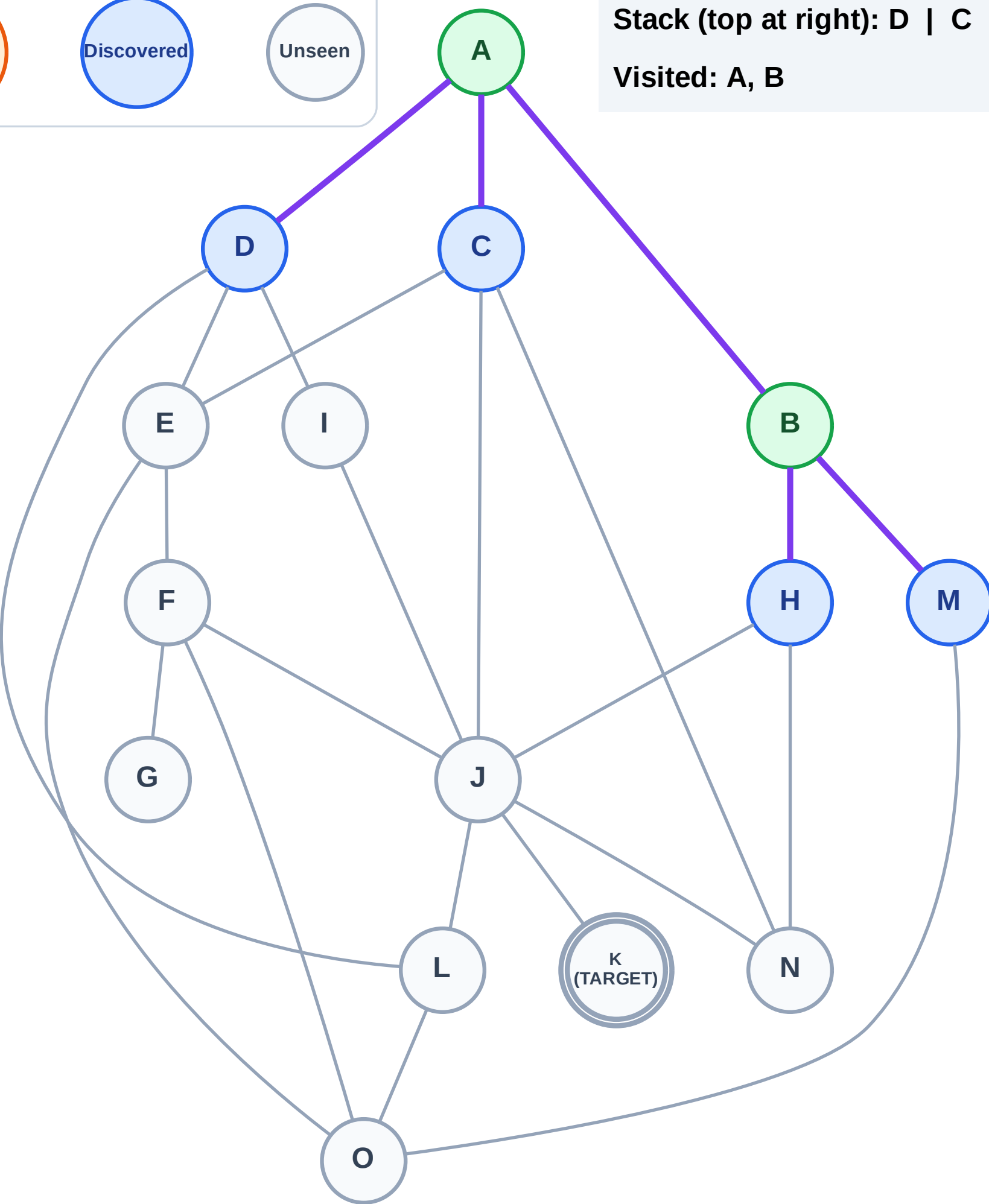
Step 5: Discover M, H from B

Legend



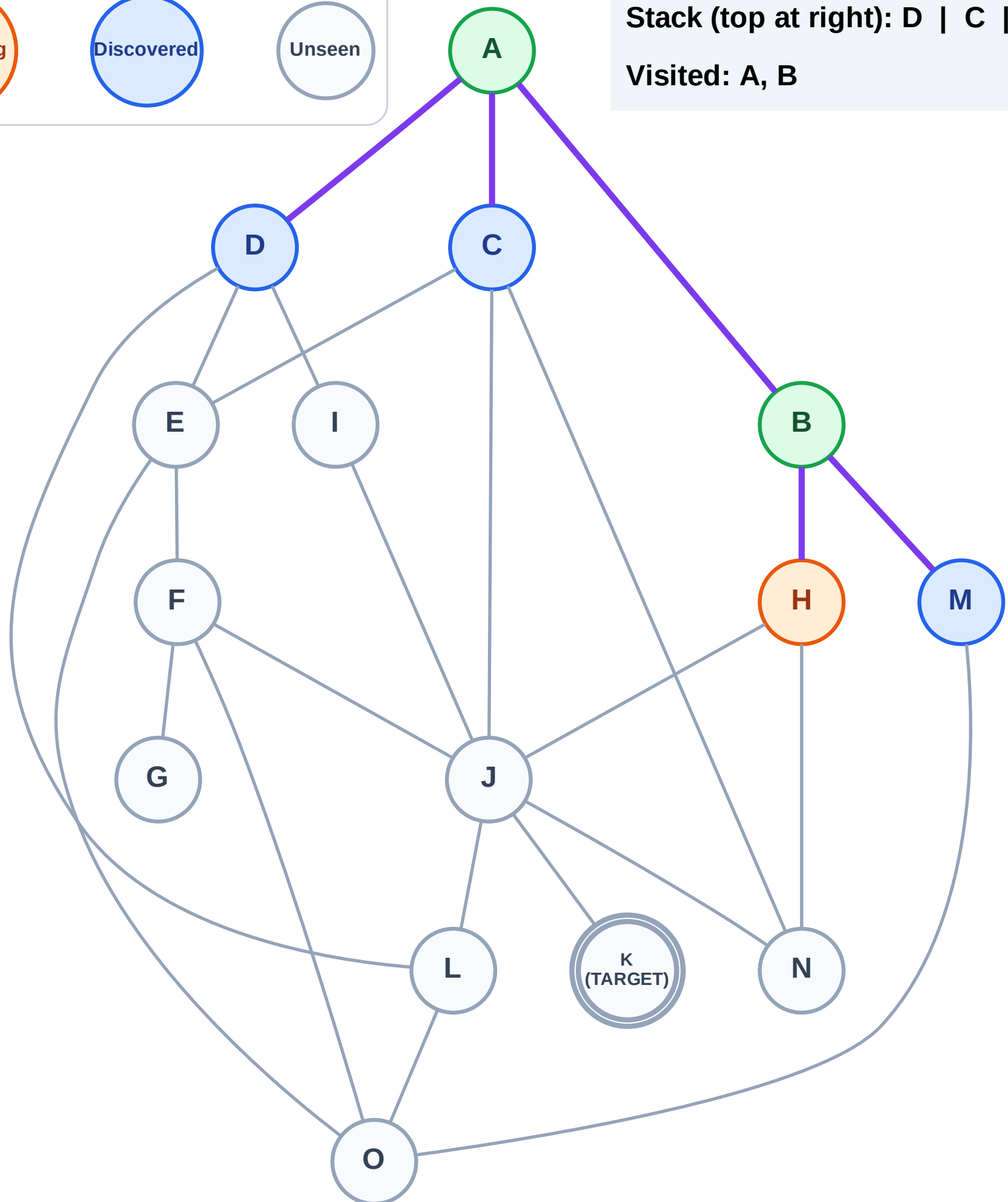
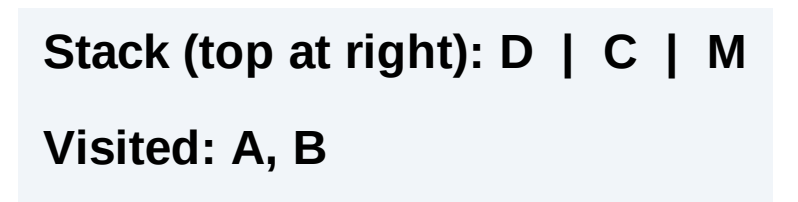
Stack (top at right): D | C | M | H

Visited: A, B



Step 6: Process node H

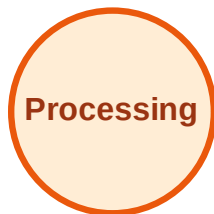
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DFS Traversal

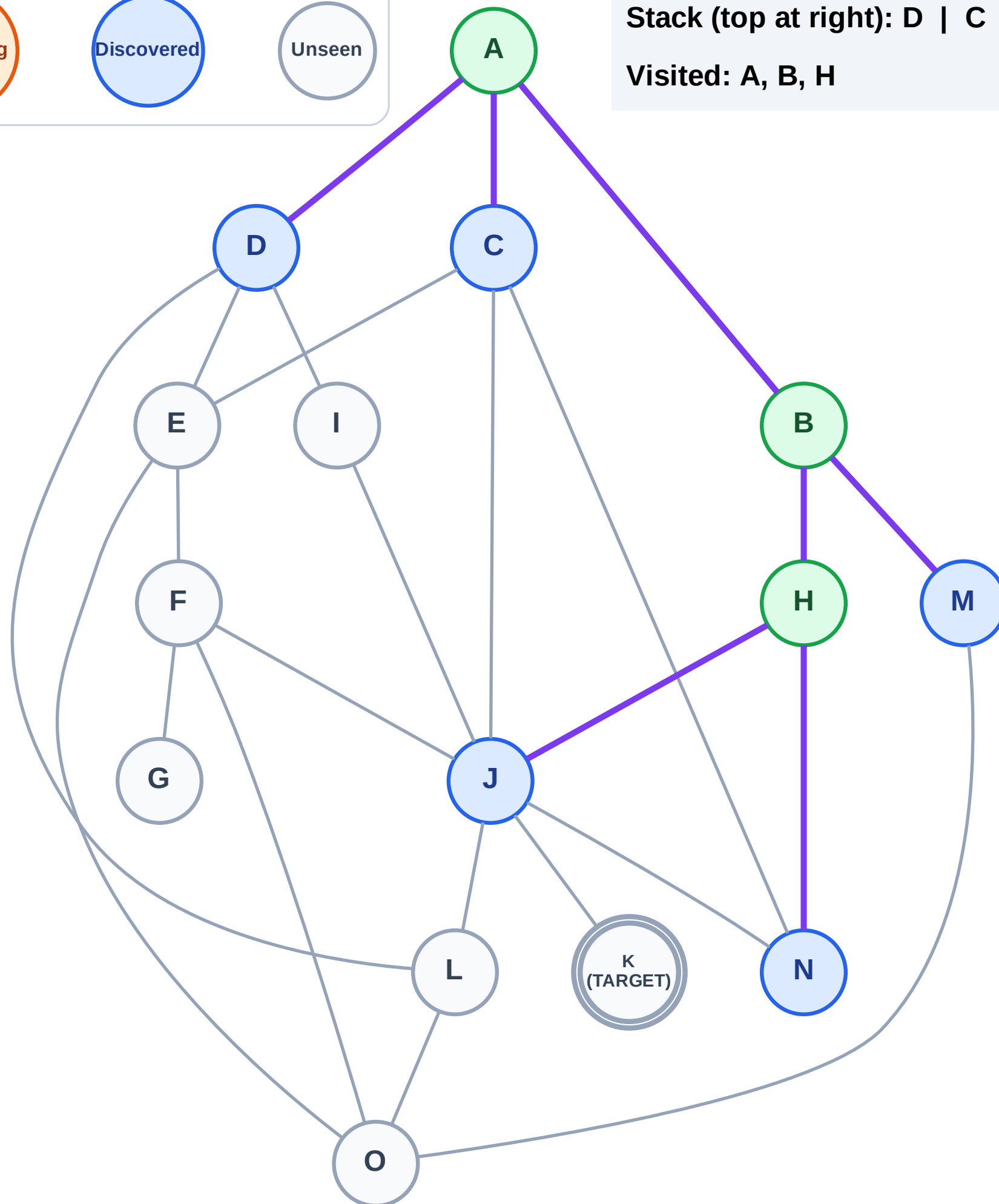
Step 7: Discover N, J from H

Legend



Stack (top at right): D | C | M | N | J

Visited: A, B, H



DFS Traversal

Step 8: Process node J

Legend

Complete

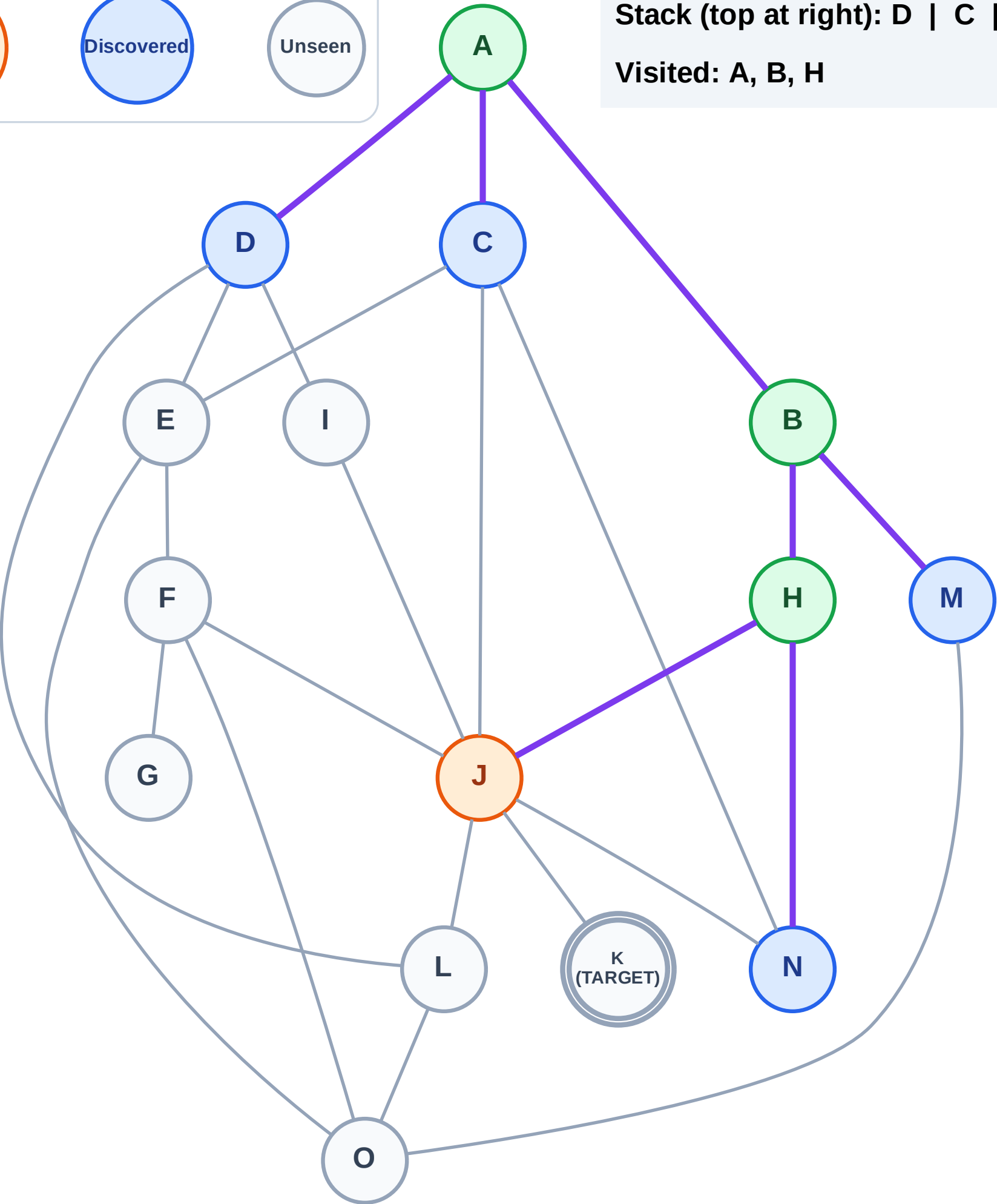
Processing

Discovered

Unseen

Stack (top at right): D | C | M | N

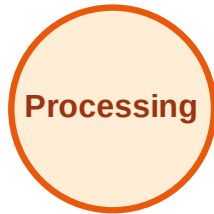
Visited: A, B, H



DFS Traversal

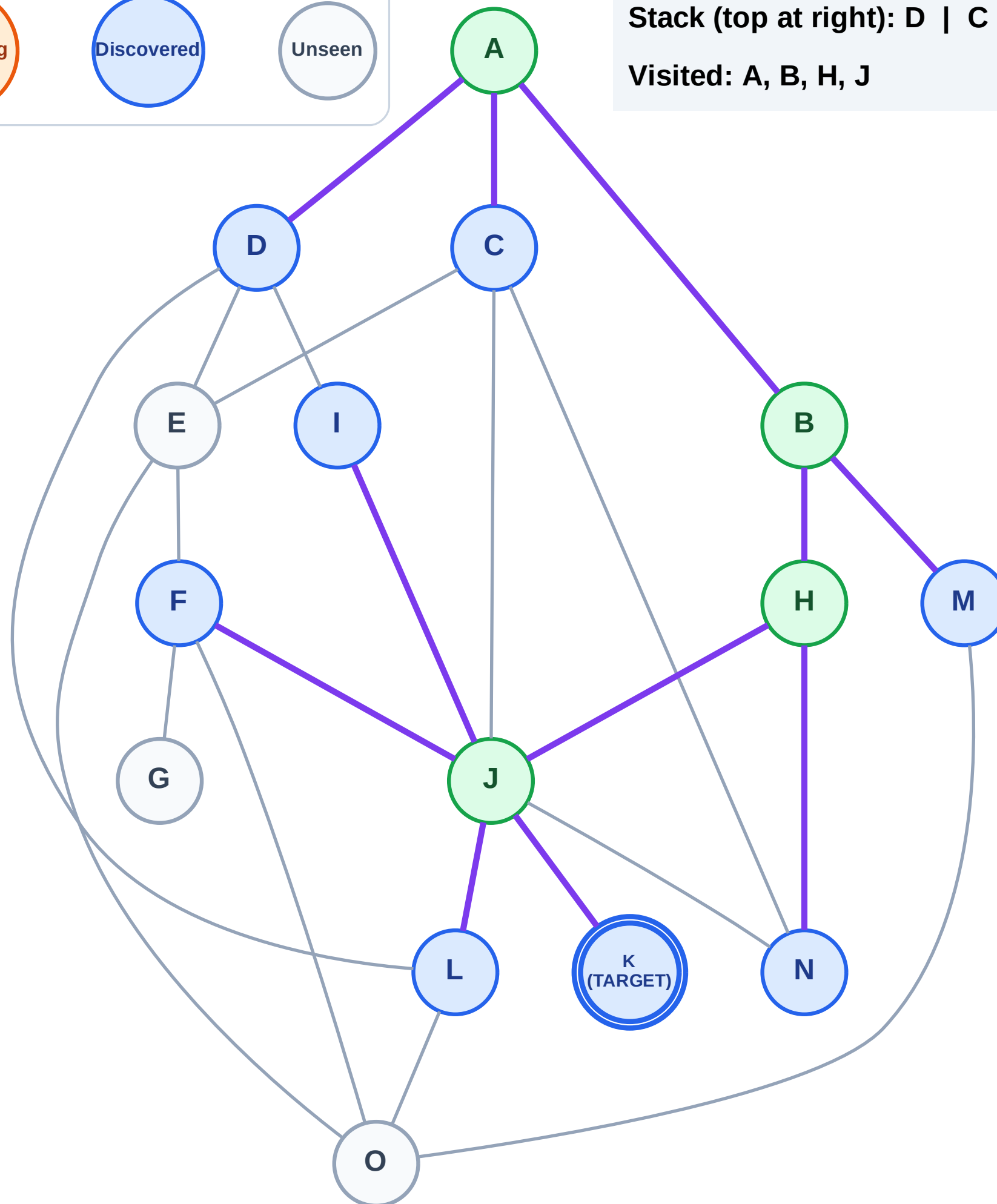
Step 9: Discover L, K, I, F from J

Legend



Stack (top at right): D | C | M | N | L | K | I | F

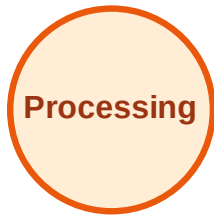
Visited: A, B, H, J



DFS Traversal

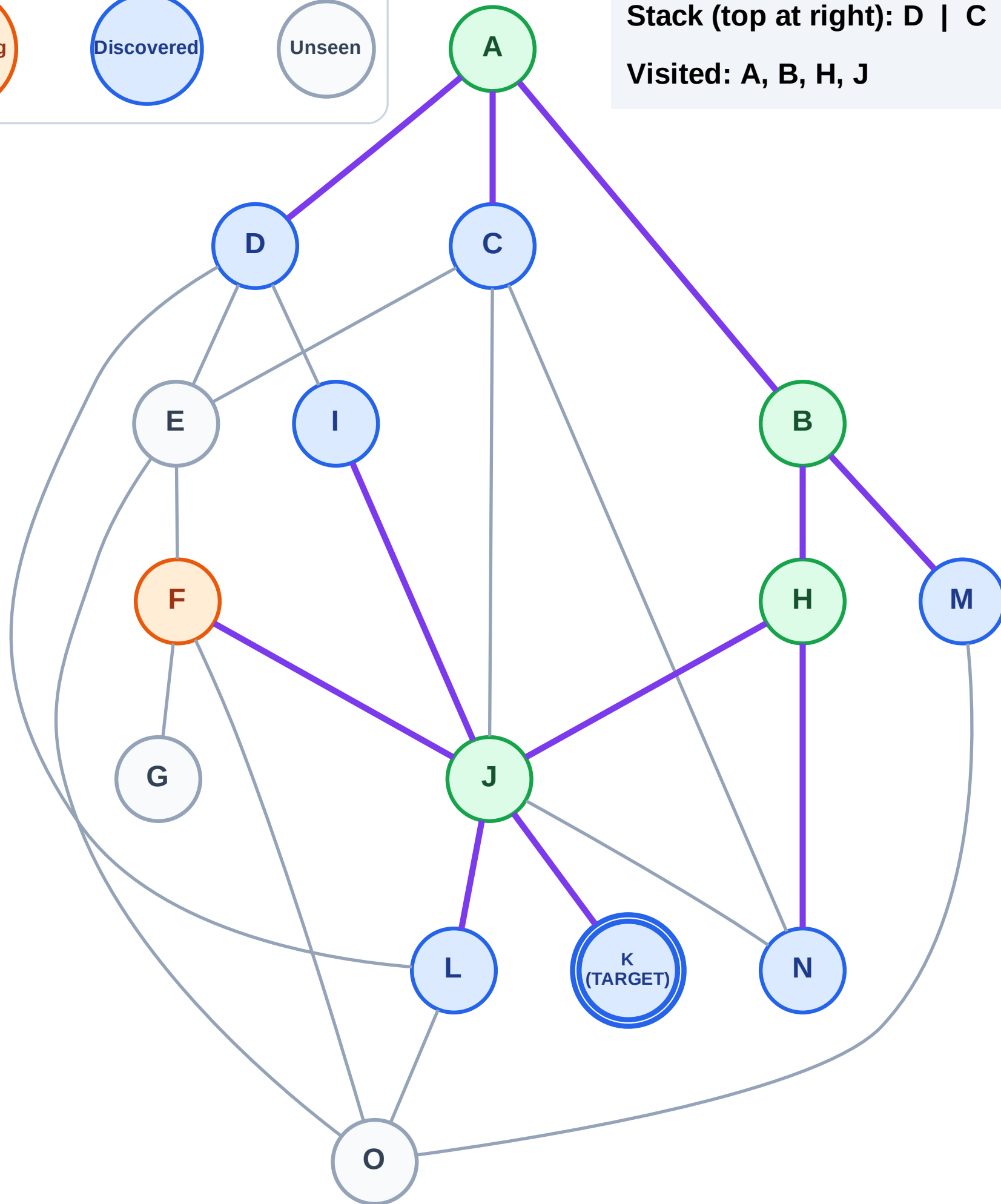
Step 10: Process node F

Legend



Stack (top at right): D | C | M | N | L | K | I

Visited: A, B, H, J



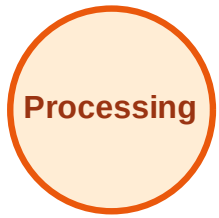
DFS Traversal

Step 11: Discover O, G, E from F

Legend



Complete



Processing



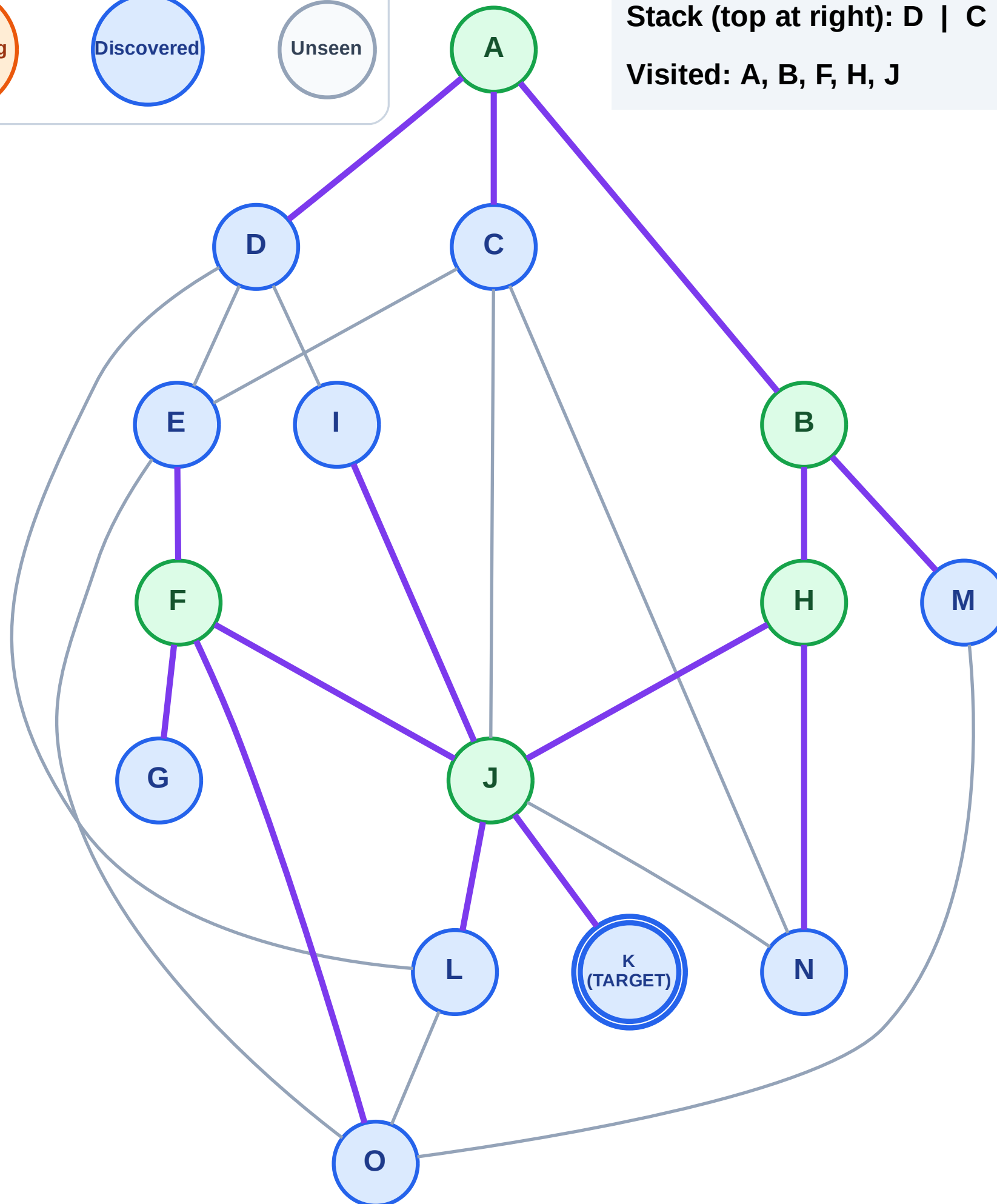
Discovered



Unseen

Stack (top at right): D | C | M | N | L | K | I | O | G | E

Visited: A, B, F, H, J



DFS Traversal

Step 12: Process node E

Legend

Complete

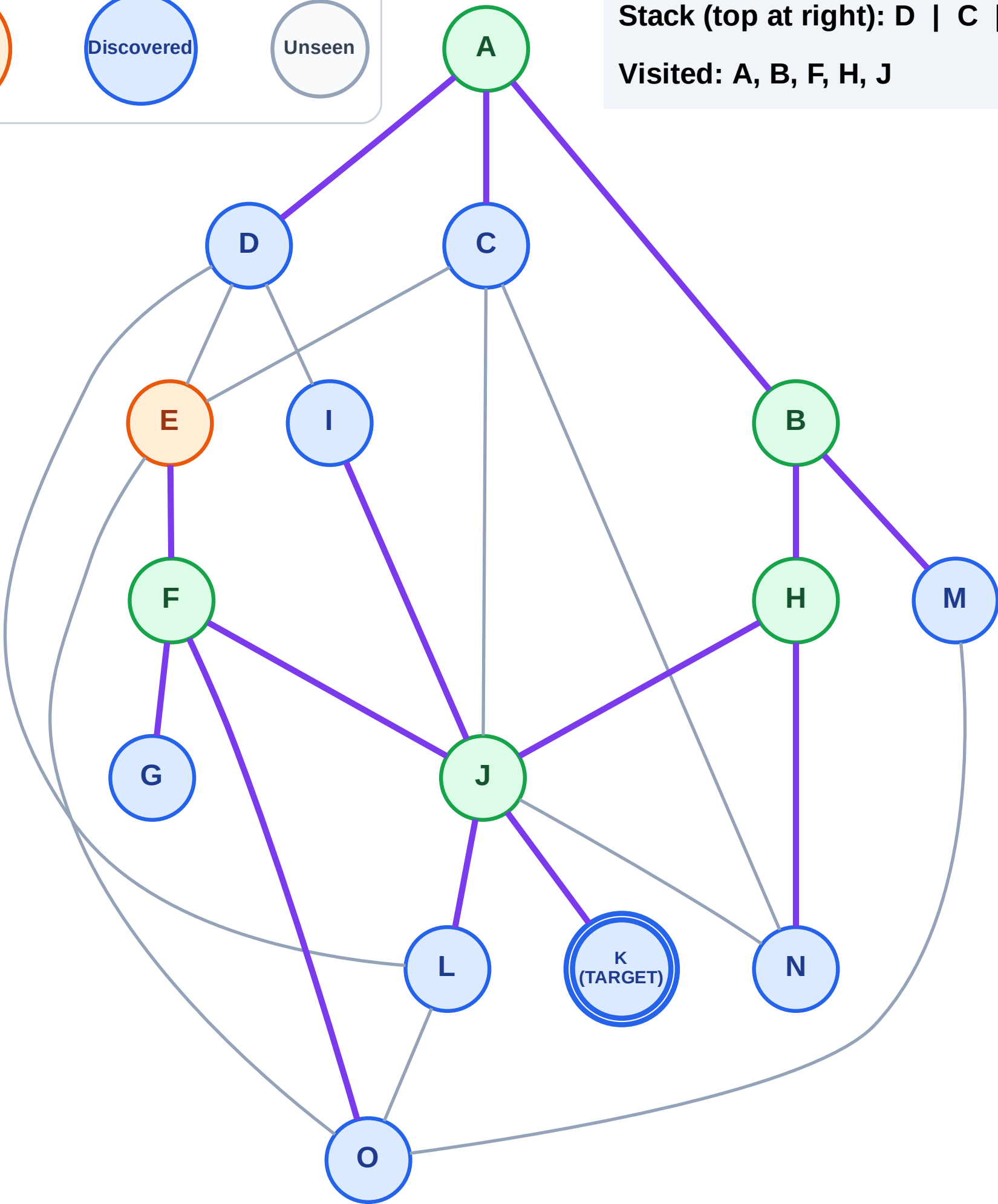
Processing

Discovered

Unseen

Stack (top at right): D | C | M | N | L | K | I | O | G

Visited: A, B, F, H, J



DFS Traversal

Step 13: No new neighbors from E

Legend

Complete

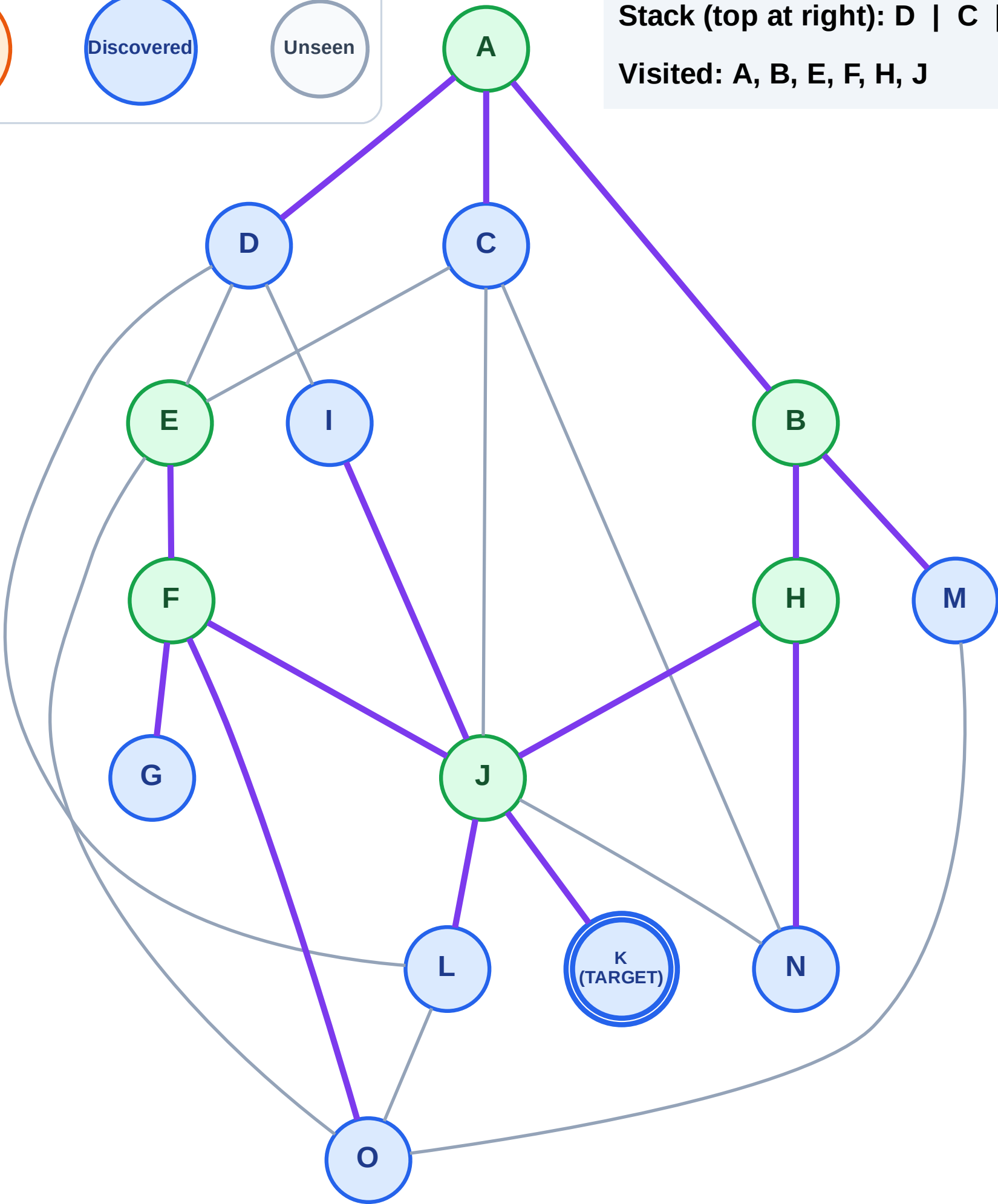
Processing

Discovered

Unseen

Stack (top at right): D | C | M | N | L | K | I | O | G

Visited: A, B, E, F, H, J



DFS Traversal

Step 14: Process node G

Legend

Complete

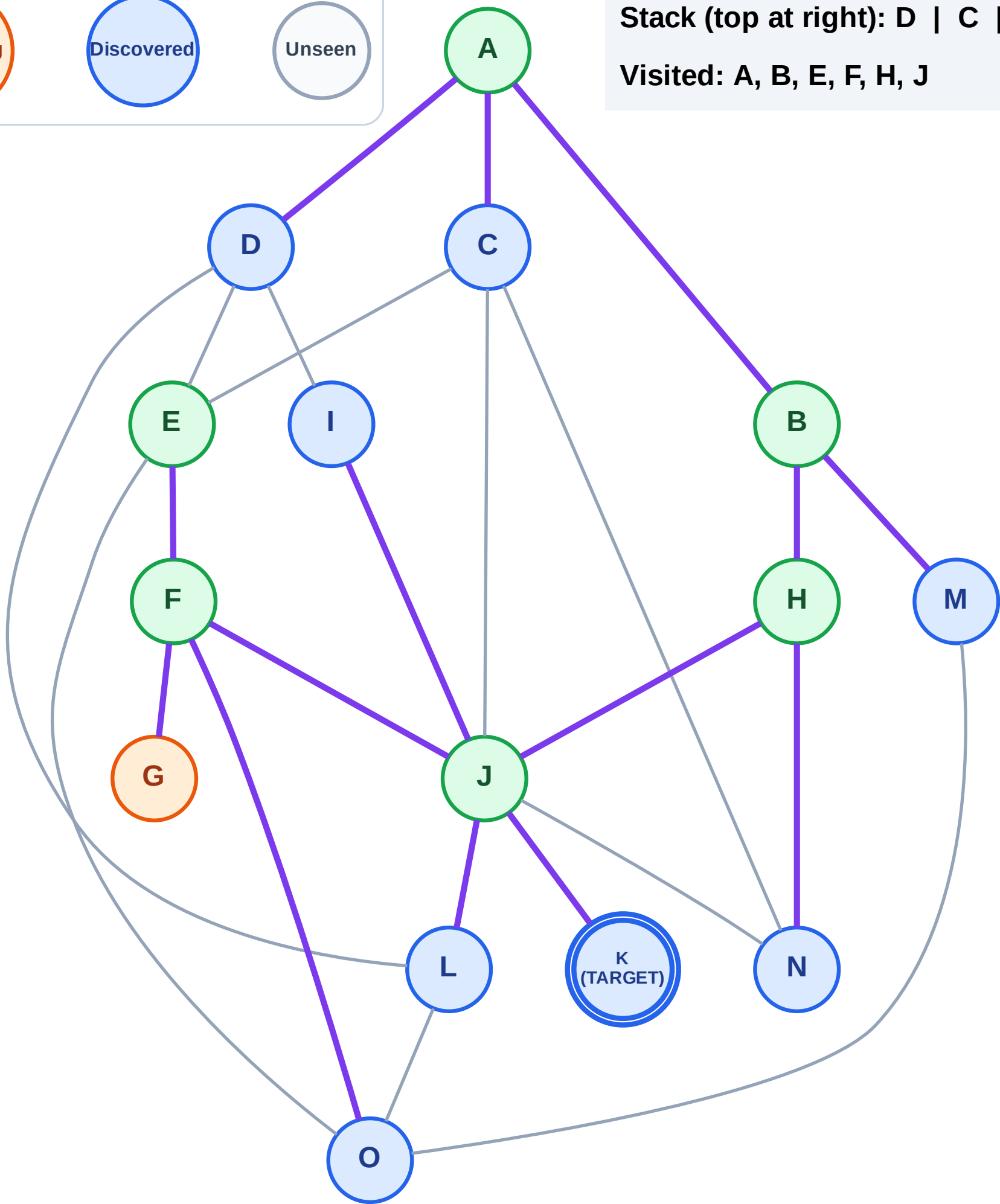
Processing

Discovered

Unseen

Stack (top at right): D | C | M | N | L | K | I | O

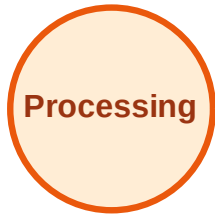
Visited: A, B, E, F, H, J



DFS Traversal

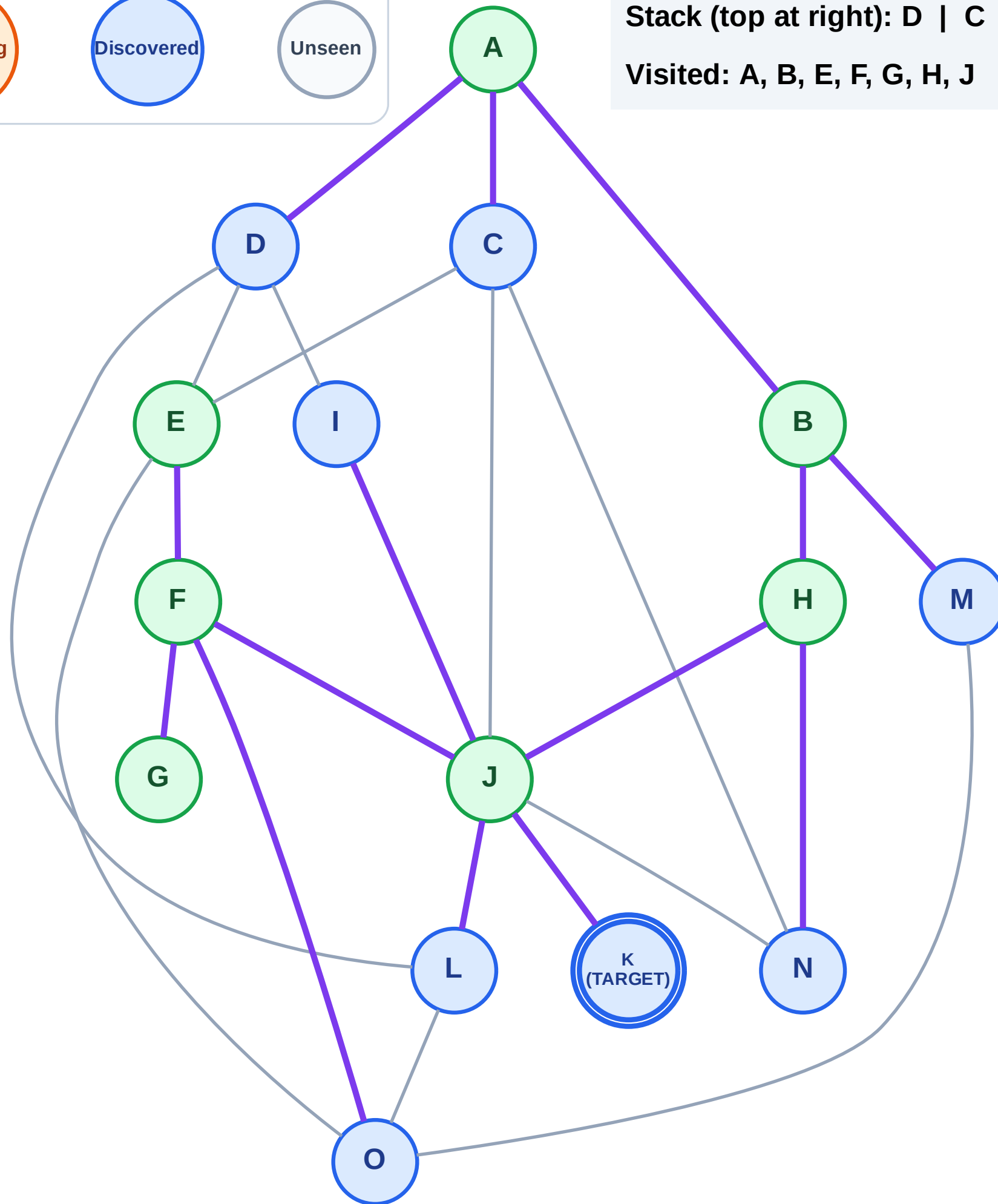
Step 15: No new neighbors from G

Legend



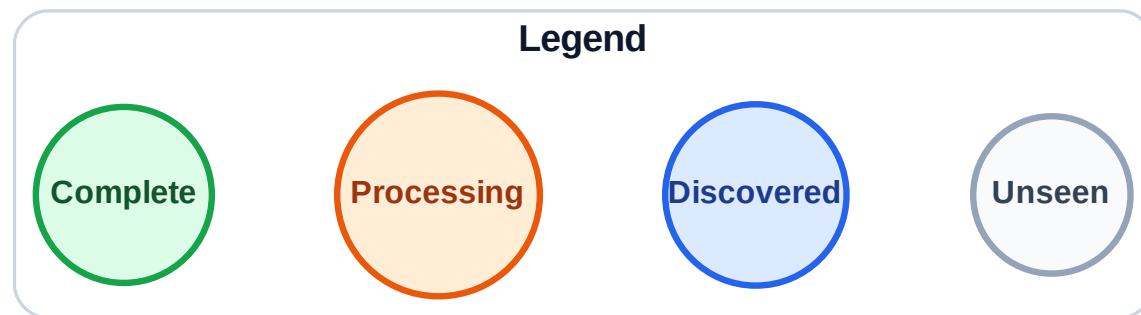
Stack (top at right): D | C | M | N | L | K | I | O

Visited: A, B, E, F, G, H, J



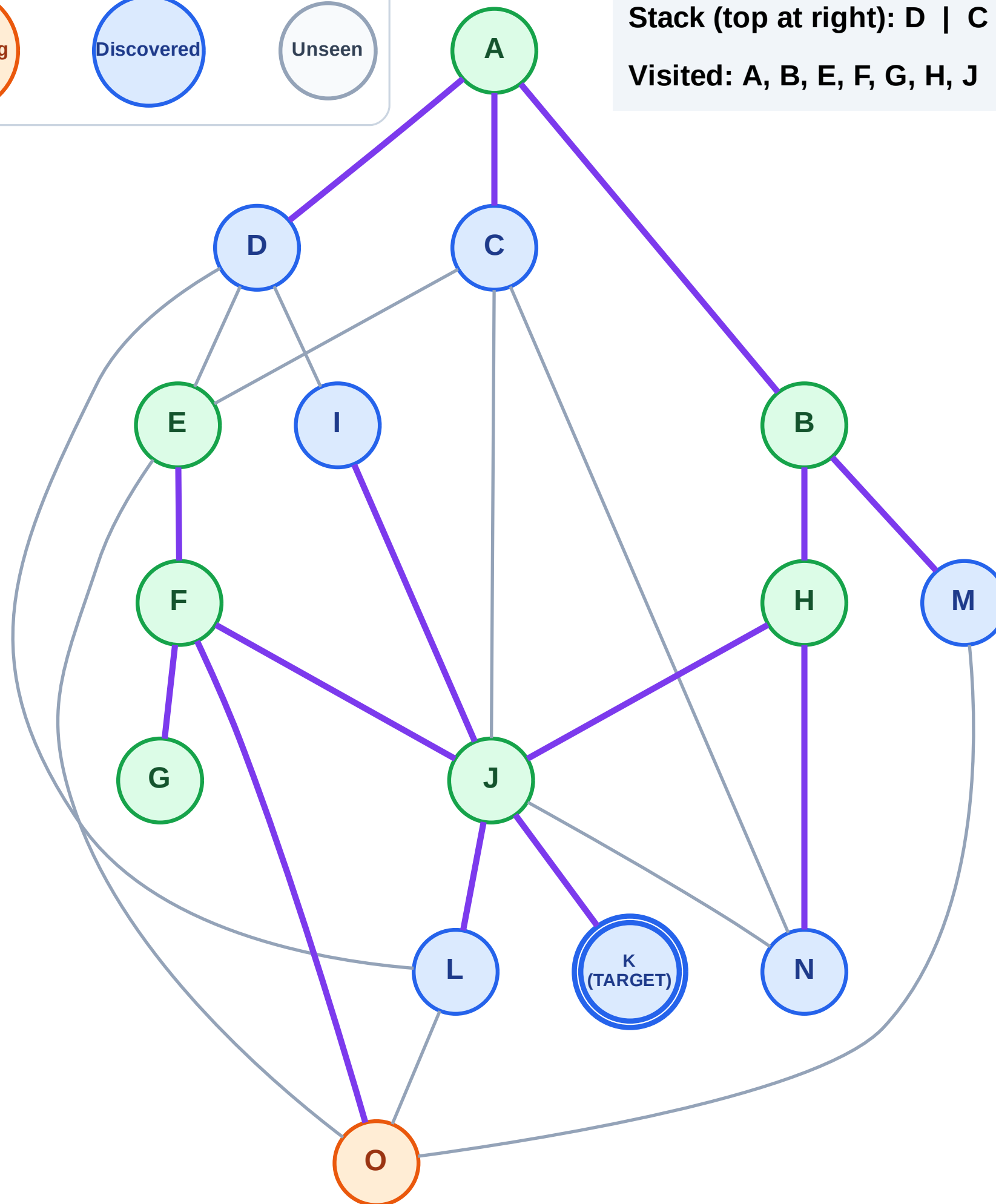
DFS Traversal

Step 16: Process node O



Stack (top at right): D | C | M | N | L | K | I

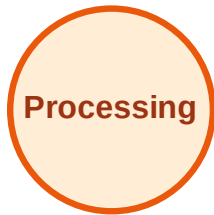
Visited: A, B, E, F, G, H, J



DFS Traversal

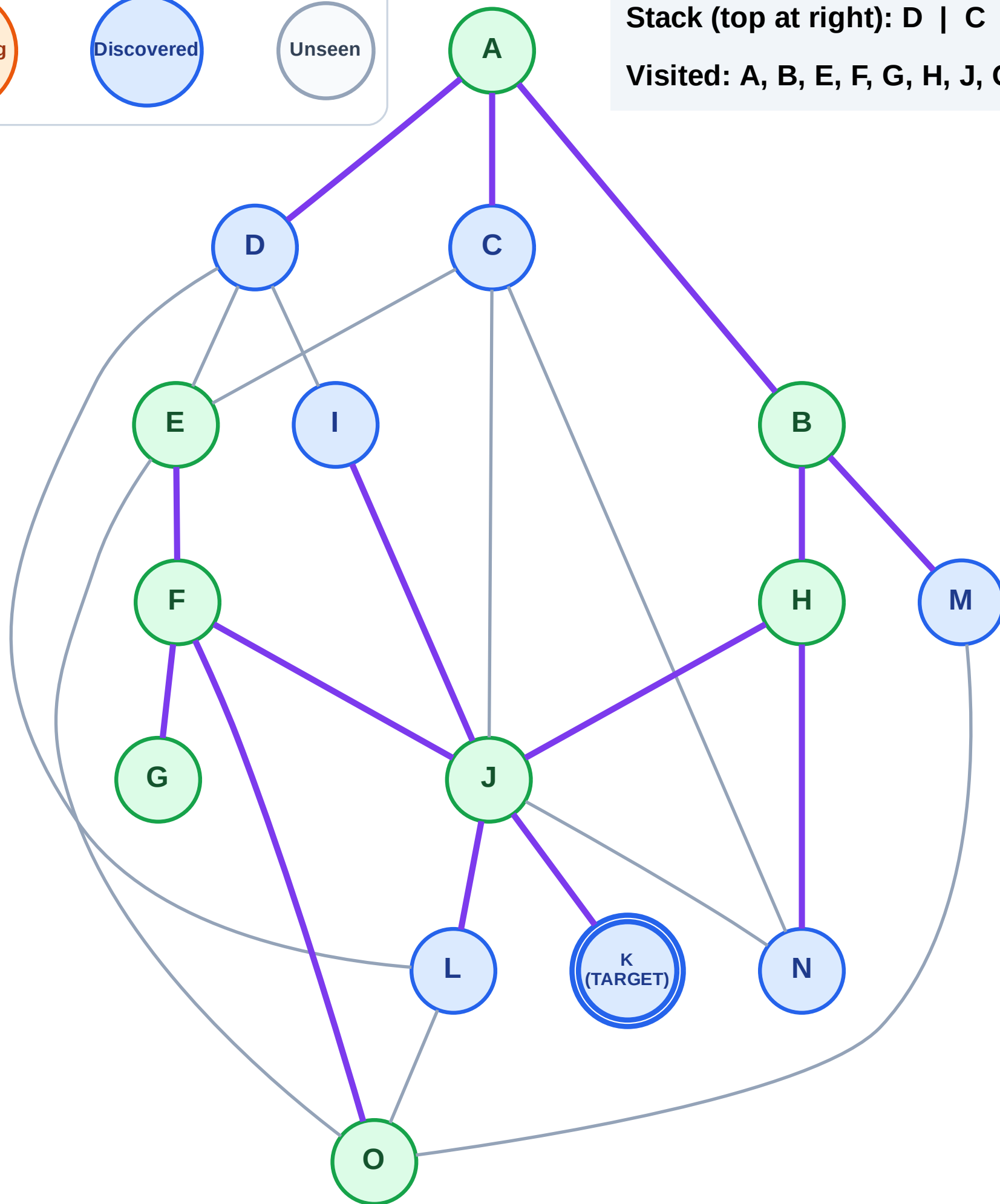
Step 17: No new neighbors from O

Legend



Stack (top at right): D | C | M | N | L | K | I

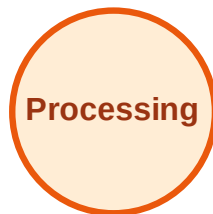
Visited: A, B, E, F, G, H, J, O



DFS Traversal

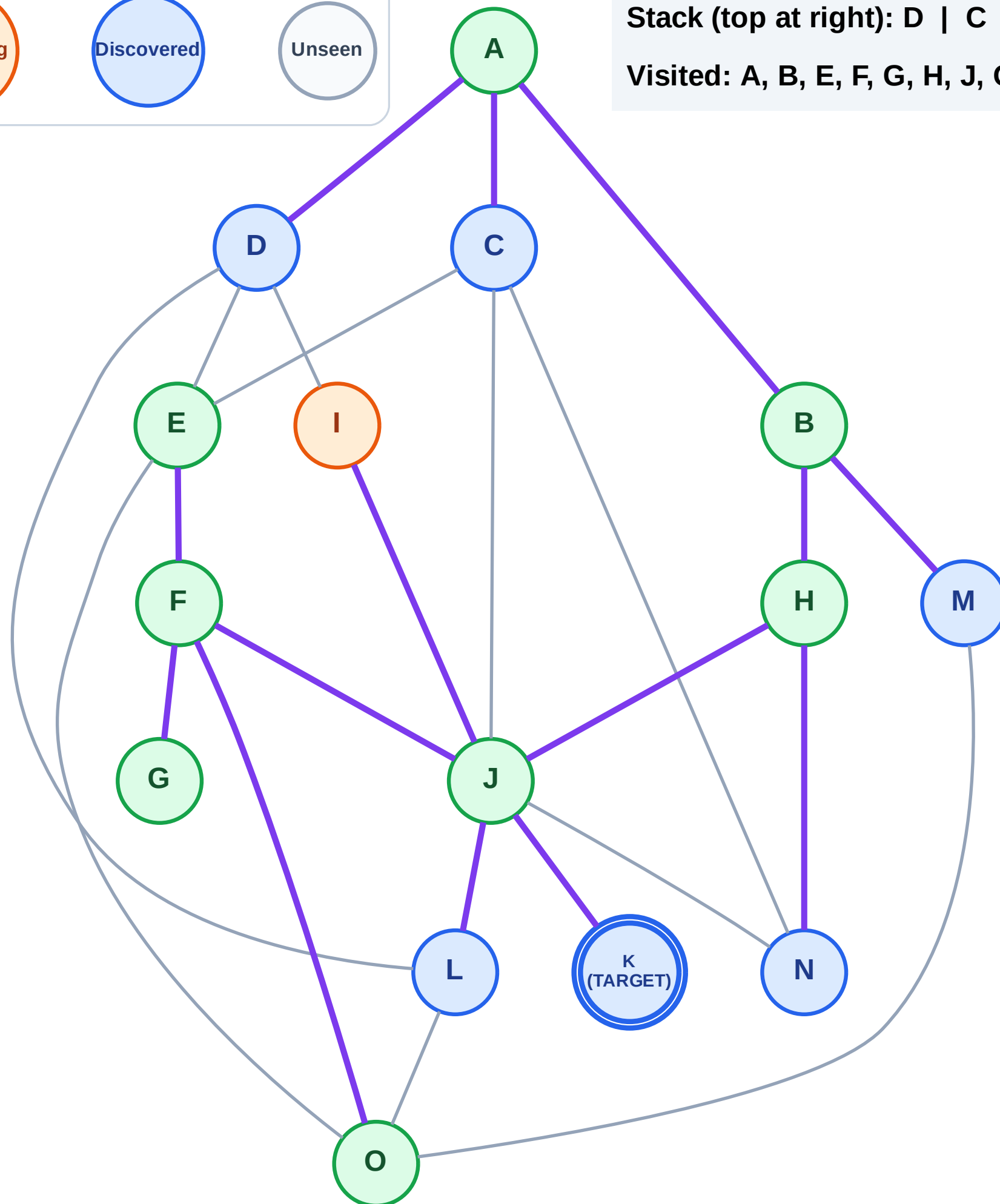
Step 18: Process node I

Legend



Stack (top at right): D | C | M | N | L | K

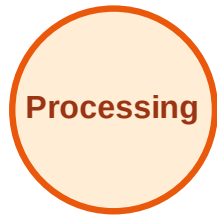
Visited: A, B, E, F, G, H, J, O



DFS Traversal

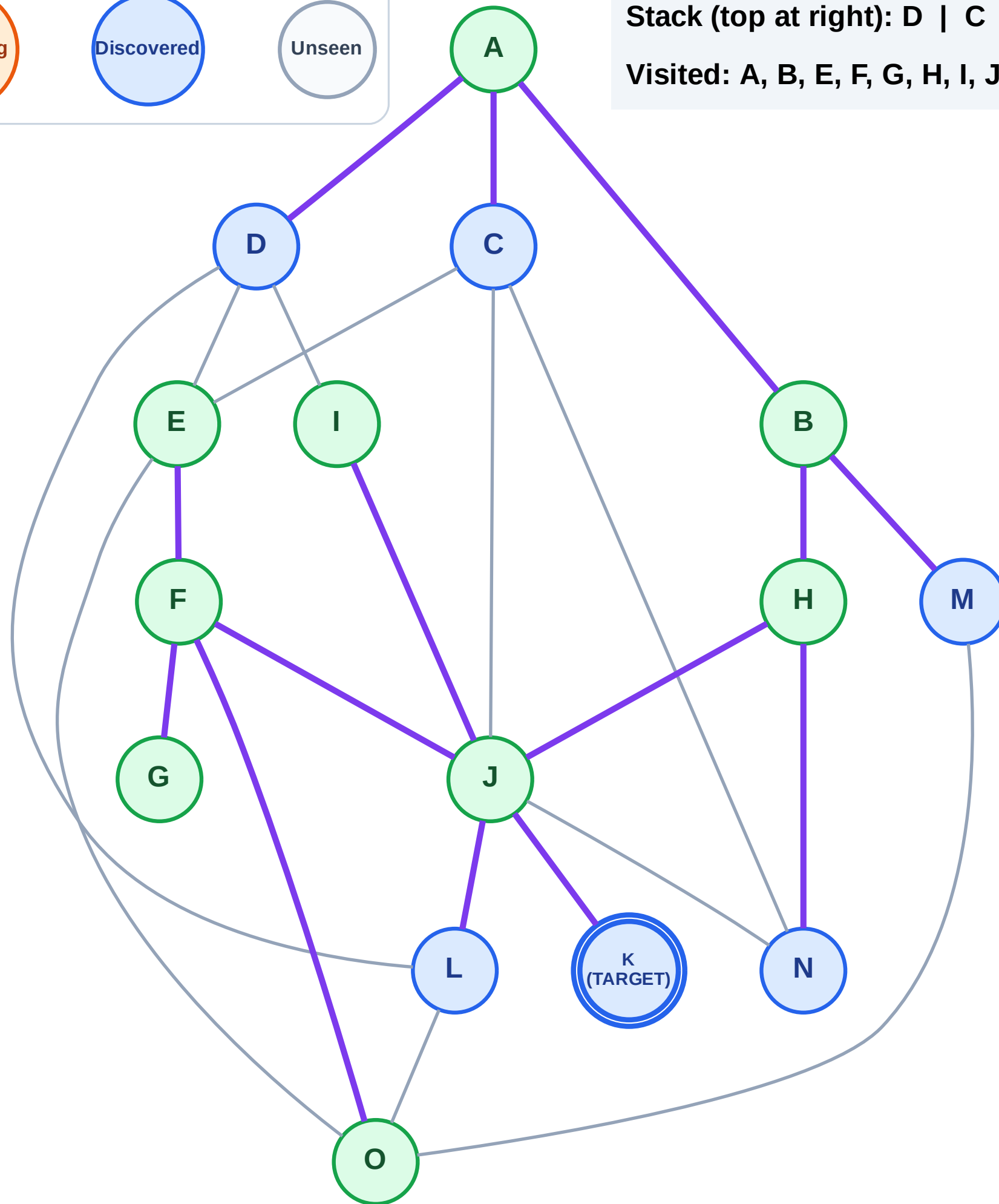
Step 19: No new neighbors from I

Legend



Stack (top at right): D | C | M | N | L | K

Visited: A, B, E, F, G, H, I, J, O



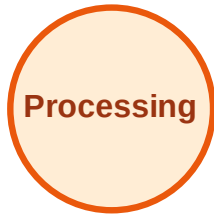
DFS Traversal

Step 20: Process node K

Legend



Complete



Processing



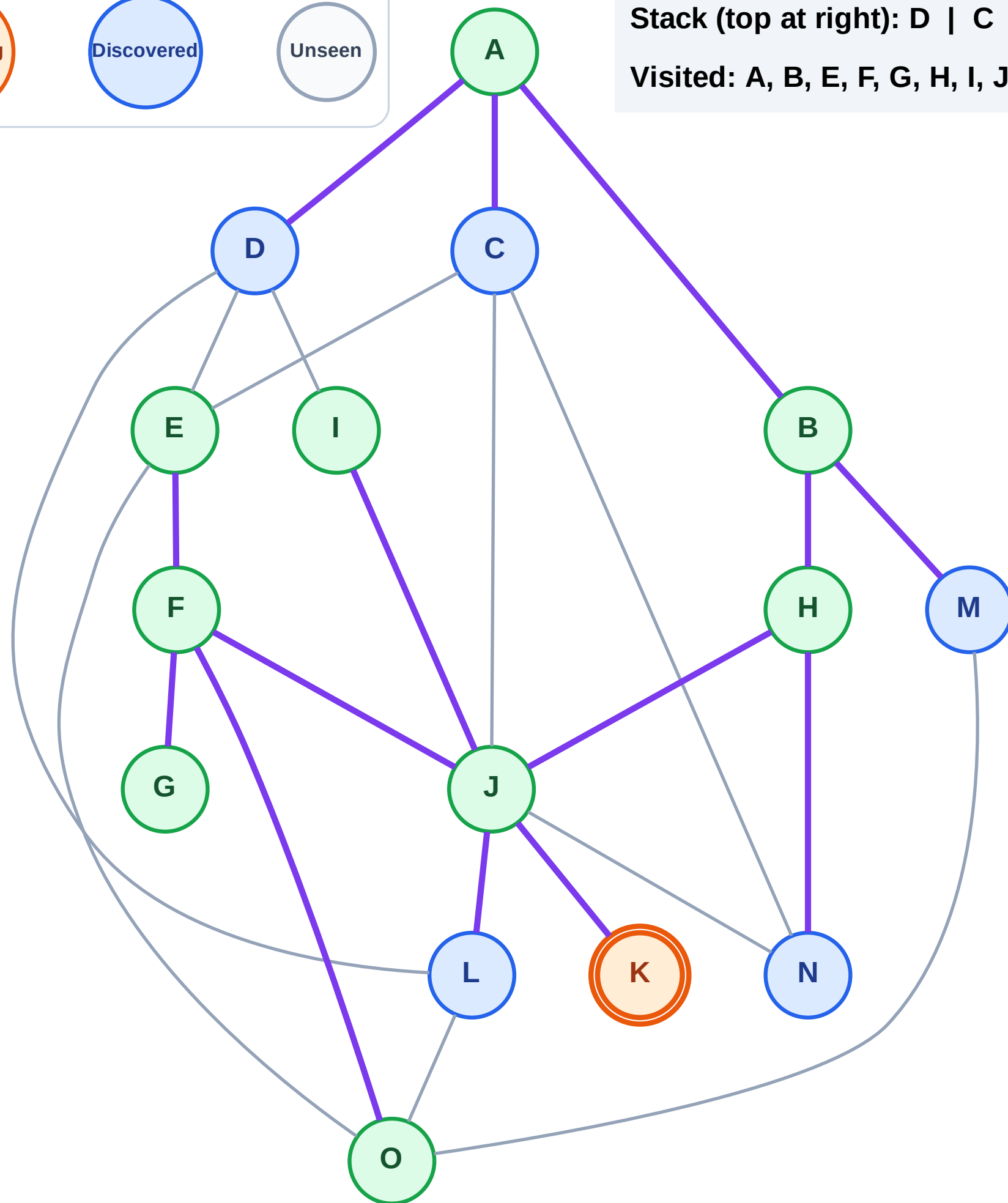
Discovered



Unseen

Stack (top at right): D | C | M | N | L

Visited: A, B, E, F, G, H, I, J, O



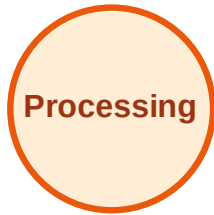
DFS Traversal

Step 21: TARGET NODE K REACHED! Path (4 hops): A → B → H → J → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): D | C | M | N | L

Visited: A, B, E, F, G, H, I, J, K, O

